

Model selection and goodness-of-fit testing: the basics

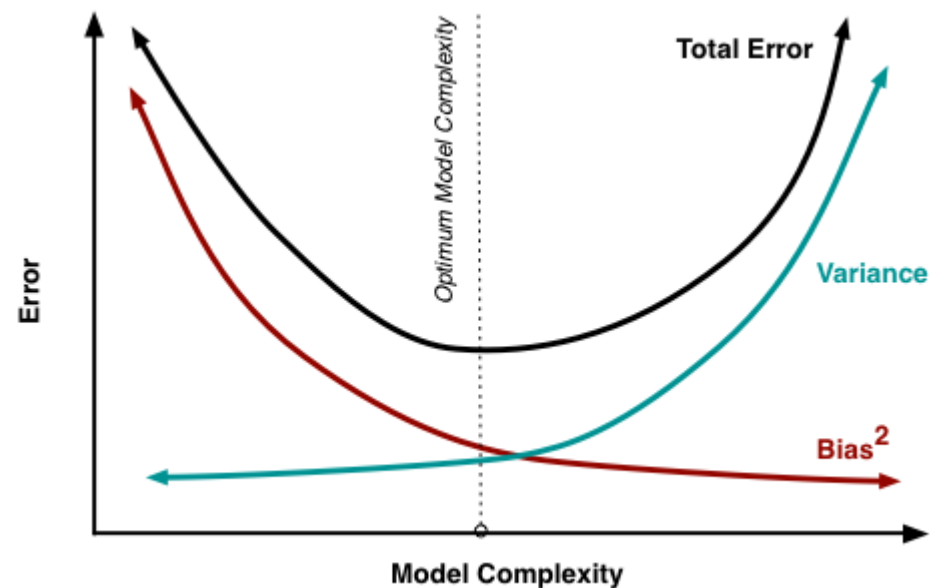
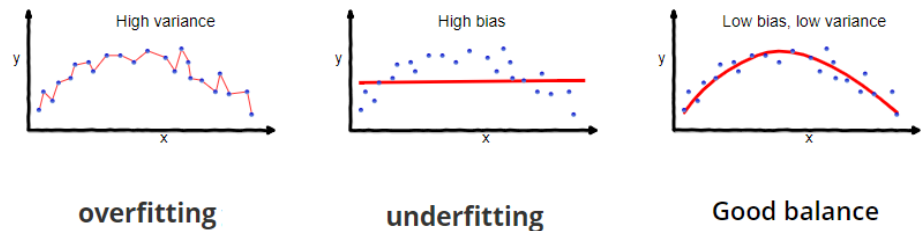


Table 1. Models examining effects of various covariates on detection probabilities of indicated breeding pairs of waterfowl in North Dakota, USA (from Pagano and Arnold 2009). I added single parameters assuming an additive effect to the base model, which included $K = 16$ parameters (8 species $\times$ 2 observers). Three of these covariates were considered biologically feasible (total ducks, vegetative cover, and cover type), 6 were not (random 5, 4, 8, and 1; not Sunday, Monday, or Wednesday; and last duck seen a mallard), and I excluded 6 additional nonsense or random variables ($\Delta AIC = 0.64\text{--}2.00$) from presentation. I evaluated all models compared to the base model using Akaike's Information Criterion (AIC), ΔAIC , and changes in model deviance (Dev).

Model	AIC	ΔAIC	K	Dev
Total ducks	4,426.71	-16.62	17	4,392.71
Random 5	4,439.95	-3.38	17	4,405.95
Random 4	4,442.20	-1.13	17	4,408.19
Vegetative cover	4,442.65	-0.68	17	4,408.65
Random 8	4,442.81	-0.52	17	4,408.80
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Base model	4,443.34	0	16	4,411.33
Not Sunday, Monday, or Wednesday	4,445.09	1.75	17	4,411.08
Cover type	4,445.25	1.92	17	4,411.25
Last duck seen a mallard	4,445.33	1.99	17	4,411.32



> dredge (no)

PPPs; BCIs; CIs; p-values; DIC; AIC; BIC; AICc; WAIC; Bayesian p-values; deviance; egads!

Announcements

1. Next week we'll start case studies
2. Thanks for submitting initial summaries, I'll review over the weekend
3. Final report formatting is extremely flexible*
 - Code and data [reproducible]
 - Conceptual | causal diagram
 - Methods section (w/results?)
 - Dissertation chapter
 - Manuscript draft

***You don't have to use SEM. If you don't, just explain why in ~1 page**

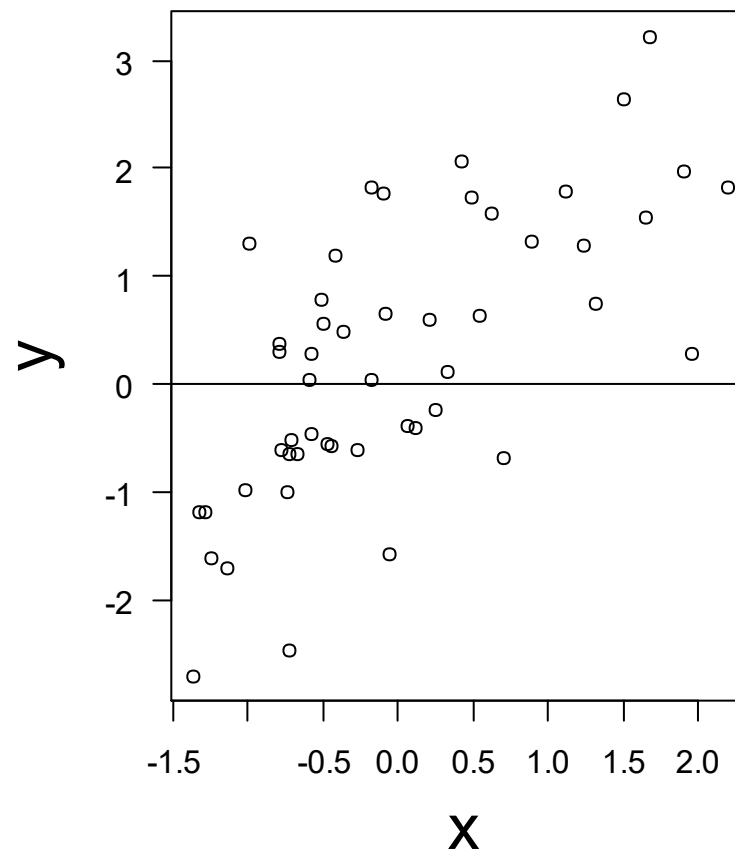
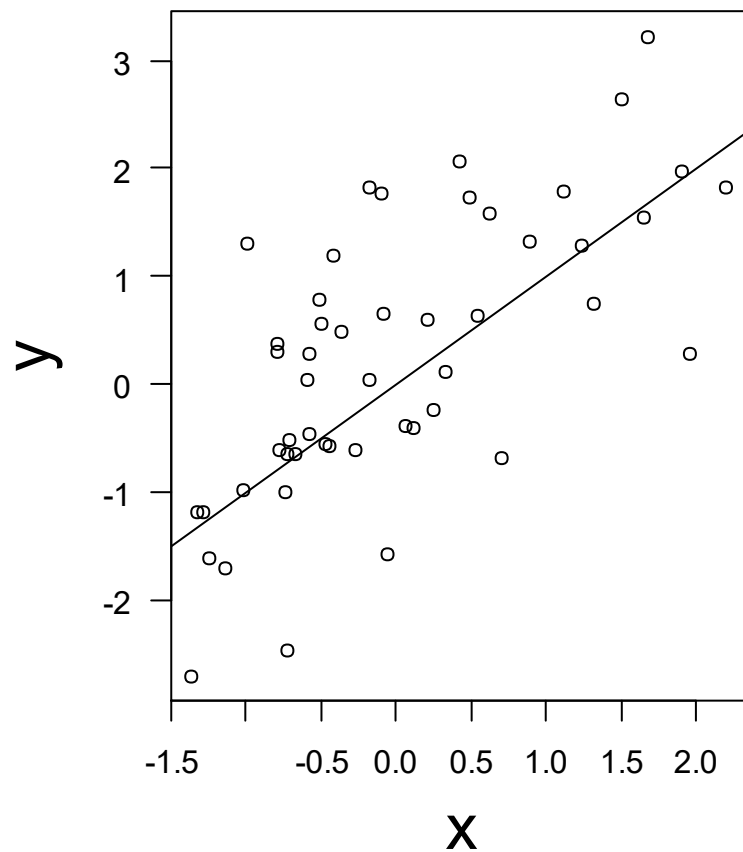
A road map for today

1. Likelihood [rapid fire]
2. Model selection [lots of examples]
3. Bayesian model selection [slightly different versions of AIC]
4. Goodness-of-fit testing

***Things will always be confusing! All models are wrong!**

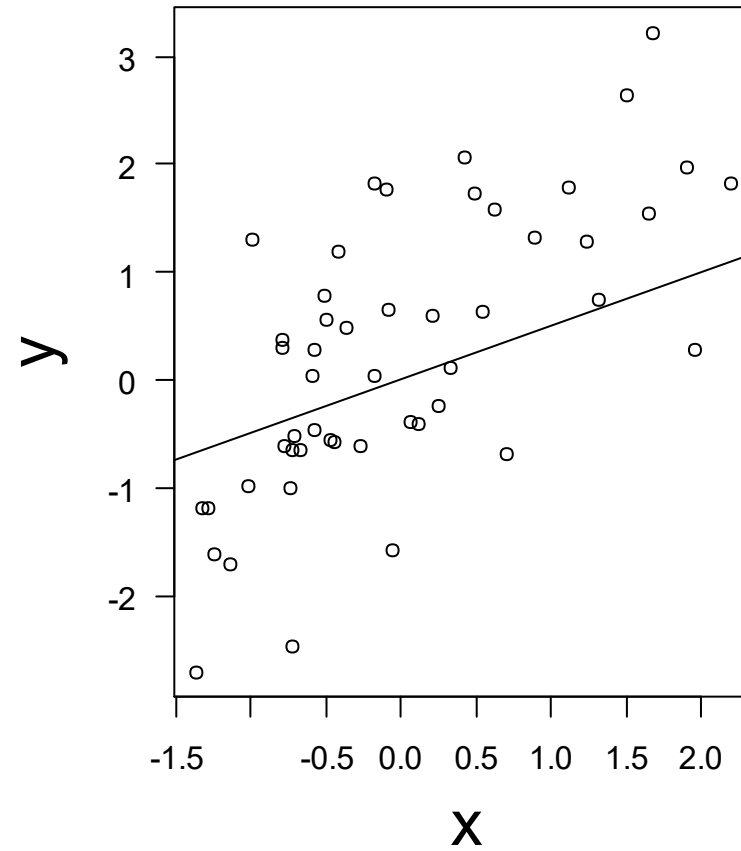
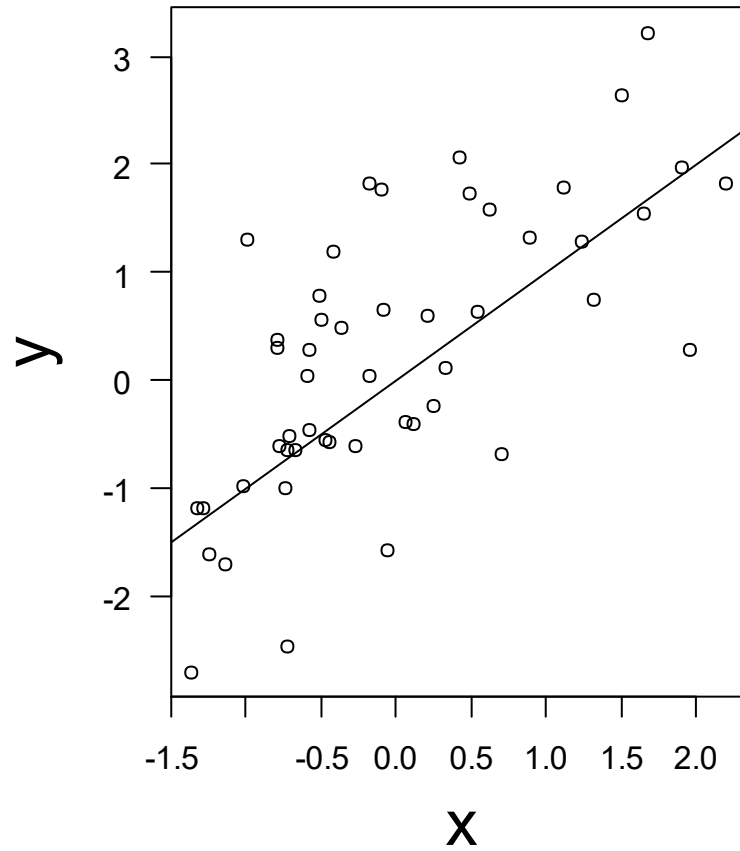
1. Likelihood: $p(y|\theta)$

Which one of these lines is more likely?



Why?

What about these lines?



Why?

What is ‘likelihood’?: It’s exactly what it sounds like...

$$l(y_i | \boldsymbol{\beta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

We use likelihood functions to estimate the **most likely parameter values** (i.e., maximum likelihood estimation)

We use likelihood functions to **compare models** (i.e., model selection)

There are different likelihood functions for different error distributions

Poisson regression log-likelihood

$$l(y_i|\boldsymbol{\beta}) = \sum_{i=1}^n y_i \times \ln(\mu_i) - \sum_{i=1}^n \mu_i - \sum_{i=1}^n \ln(y_i!)$$

Linear regression log-likelihood

$$l(y_i|\boldsymbol{\beta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

Log-likelihood function for linear regression...

$$l(y_i | \boldsymbol{\beta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

Log-likelihood function for linear regression...

$$l(y_i | \beta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

1. The data

Log-likelihood function for linear regression...

$$l(y_i | \beta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

1. The data

2. The parameter estimates and expected value ($\mu = \beta X$)

Log-likelihood function for linear regression...

$$l(y_i | \beta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

1. The data
2. The parameter estimates and expected value ($\mu = \beta X$)
3. The residual variance

Log-likelihood function for linear regression...

$$l(y_i | \beta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

1. The data
2. The parameter estimates and expected value ($\mu = \beta X$)
3. The residual variance
4. Everything else: sample size (n) and constants

Log-likelihood function for linear regression...

$$l(y_i | \beta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

Imagine we have some dataset ($n = 50$) and a log-likelihood estimate given an intercept, slope and variance...

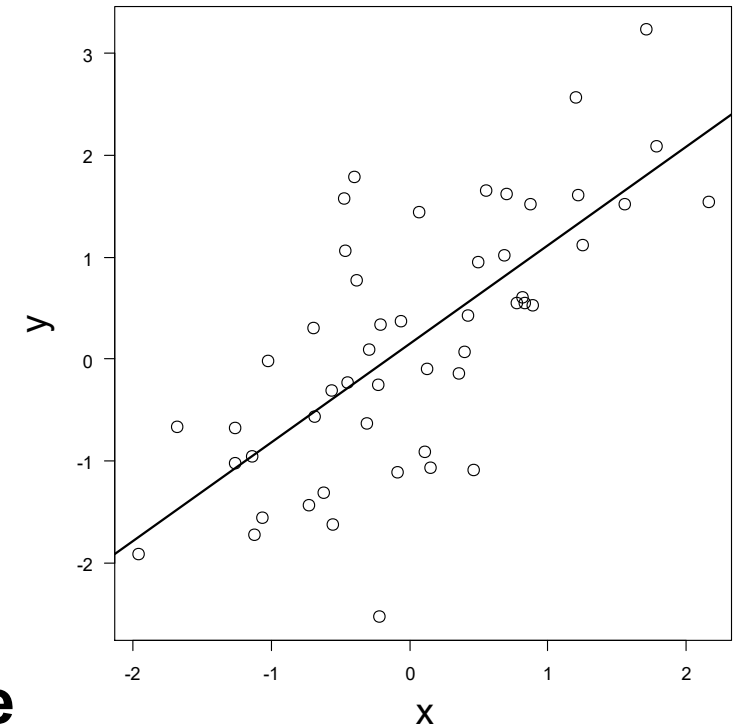
1. What will happen to the log-likelihood if we add 50 more data points?
2. What will happen to the log-likelihood if we add a covariate that explains no information of any kind ($\beta = 0$)

What will happen to the log-likelihood if we add 50 more data points?

$$l(y_i | \beta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

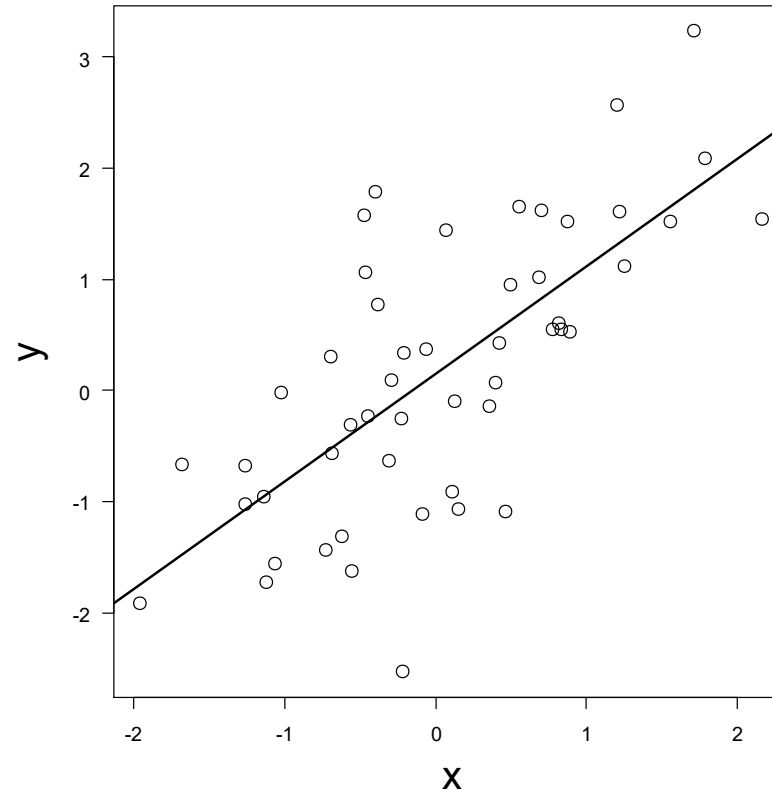
Likelihoods are relative to data

If you change the data, likelihoods are not comparable



What will happen to the log-likelihood if we add a covariate that explains no information of any kind ($\beta = 0$)?

$$l(y_i | \beta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$



2. Model selection

How do we 'score' or rank models?

- Differences ($\delta AICc$) in $AICc$ between models of < 2 indicate near equivalence
- Differences of $AICc < 7$ among models indicate that some support may exist for 'inferior' models (i.e., just because $\delta AICc$ is > 2 doesn't mean that there is no support).

Why?

Burnham and Anderson (2002)

Some rapid-fire philosophy

- **Model selection: We use model selection to rank competing hypotheses. All of those hypotheses are wrong, or at least incomplete.**
- **A model selection table is a list of incomplete and incorrect ideas.**
- **Sometimes, none of the ideas are any better than no idea at all! i.e., your 'best' model may not be very good**

Your best model could be bad

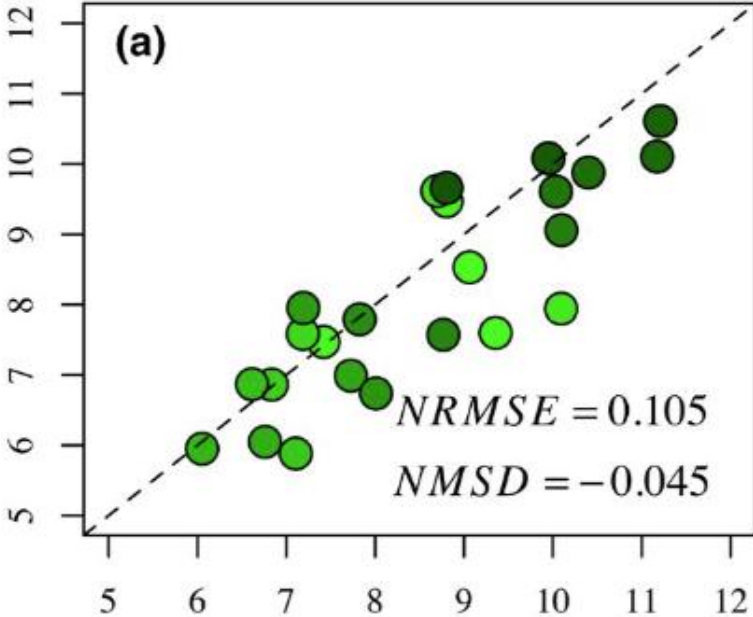
AHM balance equation

$$N_{t+1} = \gamma_s N_t \left(m S_{t,am} + (1 - m) \left(S_{t,af} + \gamma_R R_t \left(S_{t,if} + S_{t,im} \phi_f^{sum} / \phi_m^{sum} \right) \right) \right)$$

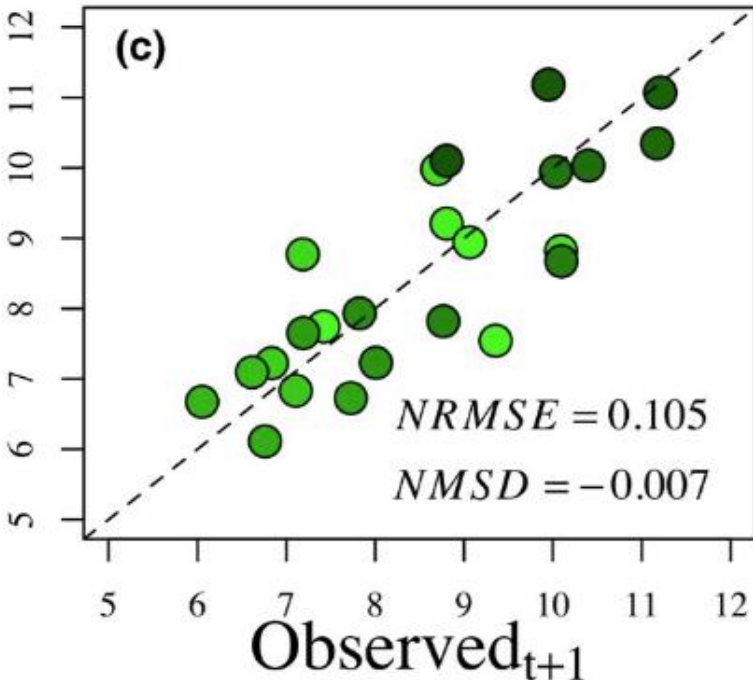
AHM submodels

Model	Features
$S_{t,age,sex} = S_{0,sex}^A (1 - K_{t,age,sex})$	Deterministic model for age- and sex-specific additive harvest mortality (S_a)
$S_{t,age,sex} = \begin{cases} S_{0,sex}^C & \text{if } K_{t,age,sex} \leq 1 - S_{0,sex}^C \\ 1 - K_{t,age,sex} & \text{if } K_{t,age,sex} > 1 - S_{0,sex}^C \end{cases}$	Deterministic model for age- and sex-specific compensatory harvest mortality (S_c) according to a segmented threshold, whereby harvest mortality becomes additive once it exceeds baseline mortality in the absence of harvest; neither of the AHM survival models account for possible effects of density dependence, habitat conditions, etc.
$R_t = 0.7166 + 0.1083W_t - 0.0373N_t$	Weak density-dependent model for reproduction (R_w) that also includes an effect of wetland availability
$R_t = 1.1390 + 0.1376W_t - 0.1131N_t$	Strong density-dependent model for reproduction (R_s); differences between the R_w and R_s models were based on the 80% confidence ellipsoid of a best-fitting linear model to historical reproduction data, and the two points on this ellipsoid with the largest and smallest values for the effect of N_t were used to develop the R_w and R_s parameters

Forecast_{t+1}



Forecast_{t+1}



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WORKING HYPOTHESIS

Ecology and Evolution
WILEY

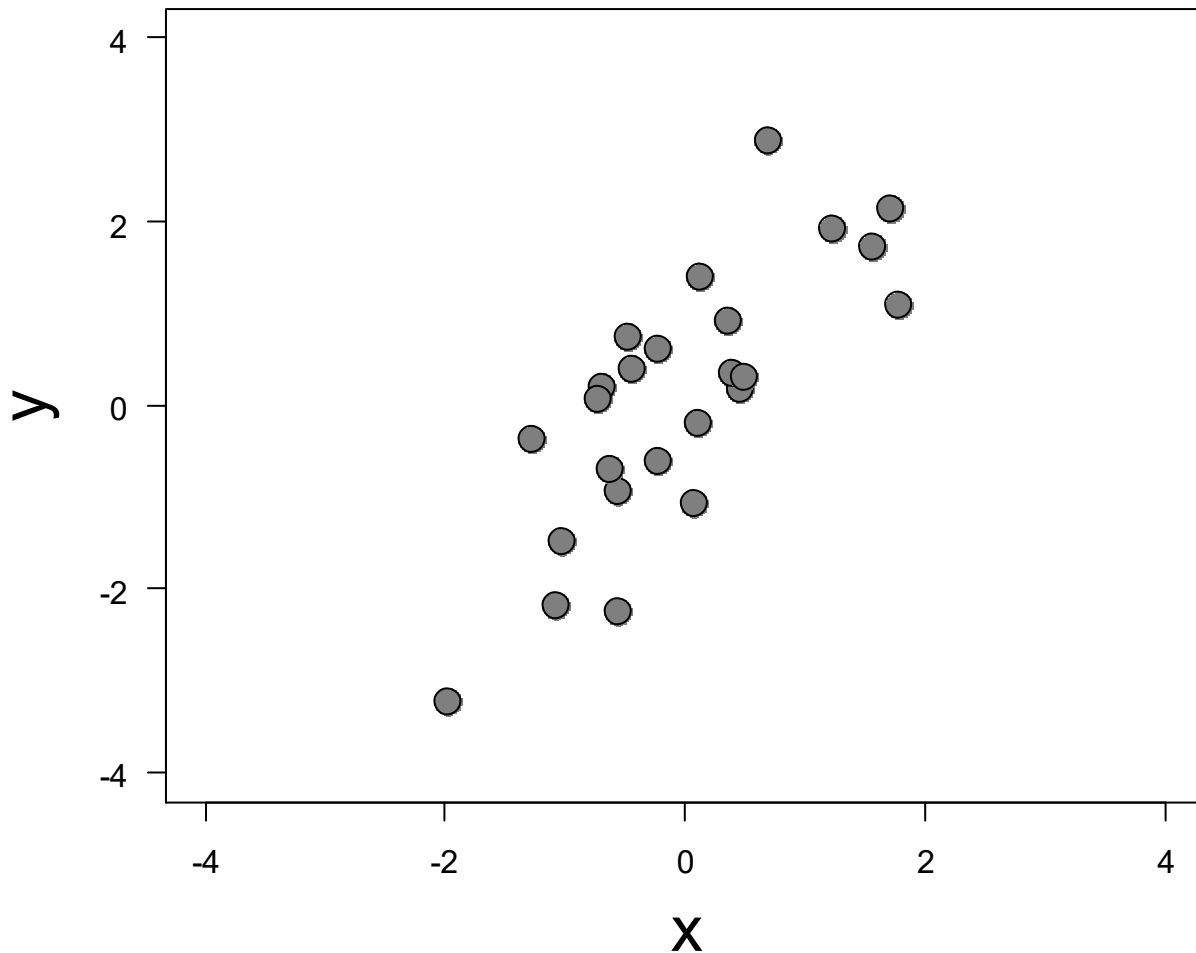
A niche for null models in adaptive resource management

David N. Koons¹ | Thomas V. Riecke^{2,3} | G. Scott Boomer⁴ | Benjamin S. Sedinger⁵ |
James S. Sedinger² | Perry J. Williams² | Todd W. Arnold⁶

$$N_{t+1} = N_t$$

Let's work through an example

Simulating data (3 parameters)



Data simulation

$$\mathbf{x} \sim \text{Normal}(0, 1)$$

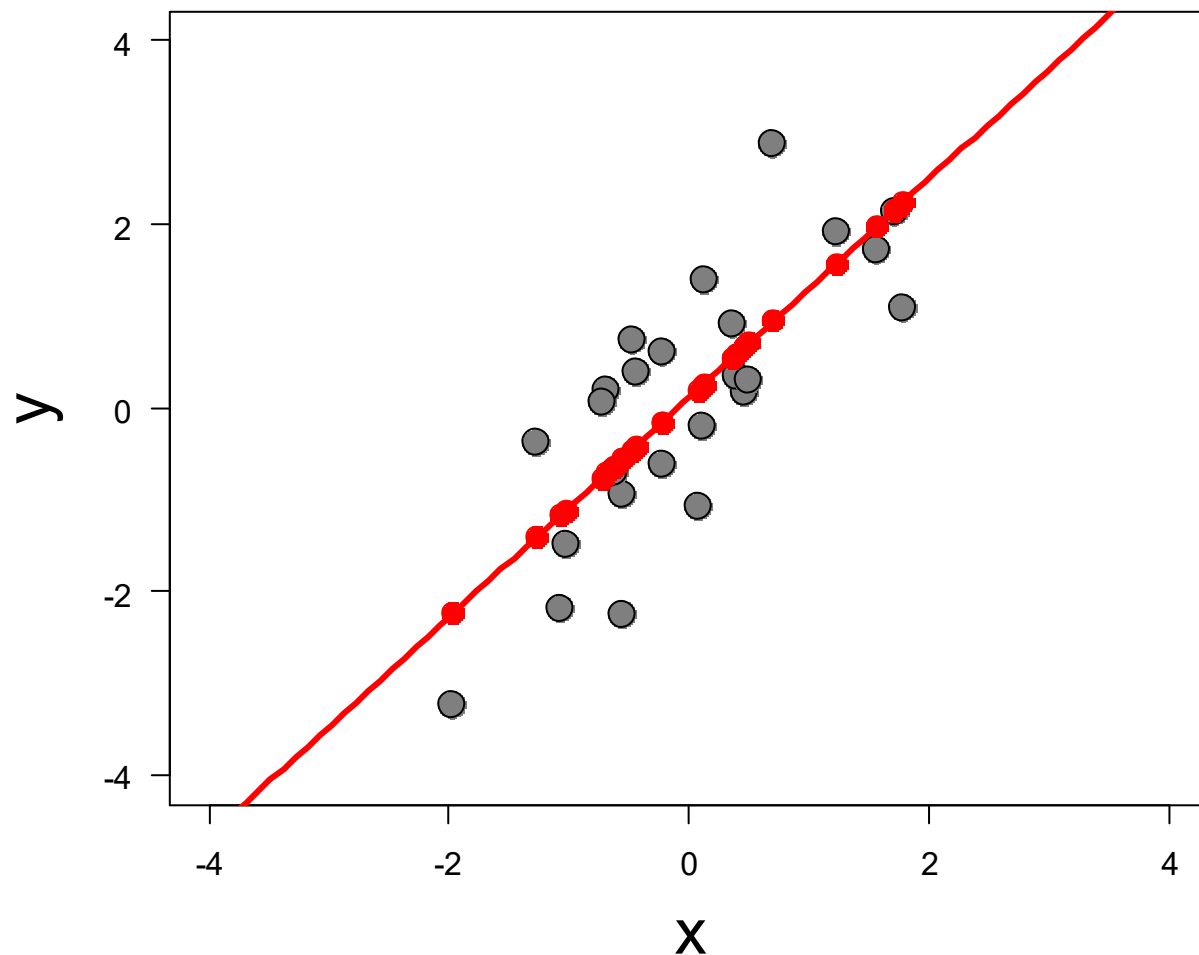
$$\mathbf{y} \sim \text{Normal}(\beta_0 + \beta_1 \mathbf{x}, \sigma^2)$$

$$\beta_0 = 0$$

$$\beta_1 = 1$$

$$\sigma = \sigma^2 = 1$$

E(y)?



Data simulation

$$\mathbf{x} \sim \text{Normal}(0, 1)$$

$$\mathbf{y} \sim \text{Normal}(\beta_0 + \beta_1 \mathbf{x}, \sigma^2)$$

$$\beta_0 = 0$$

$$\beta_1 = 1$$

$$\sigma = \sigma^2 = 1$$

Expected values

$$E(\mathbf{y}) = \boldsymbol{\eta} = \hat{\beta}_0 + \hat{\beta}_1 \mathbf{x}$$

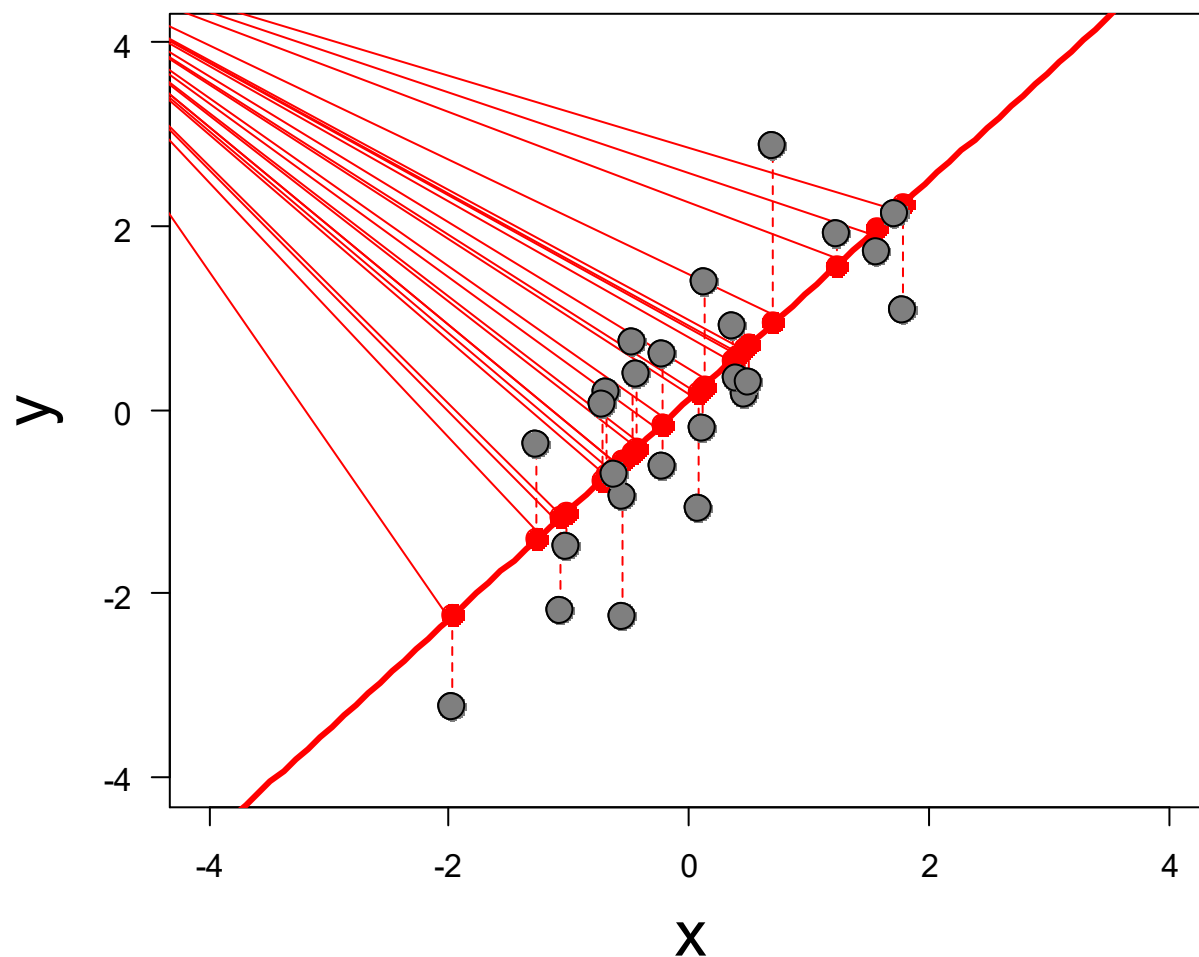
$$\hat{\beta}_0 = 0.108$$

$$\hat{\beta}_1 = 1.194$$

Expected value of y

$$l(y_i|\boldsymbol{\beta}) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\left(\sum_{i=1}^n (y_i - \mu_i)^2\right)$$

What are residuals?



Data simulation

$$\mathbf{x} \sim \text{Normal}(0, 1)$$

$$\mathbf{y} \sim \text{Normal}(\beta_0 + \beta_1 \mathbf{x}, \sigma^2)$$

$$\beta_0 = 0$$

$$\beta_1 = 1$$

$$\sigma = \sigma^2 = 1$$

Expected values

$$E(\mathbf{y}) = \boldsymbol{\eta}$$

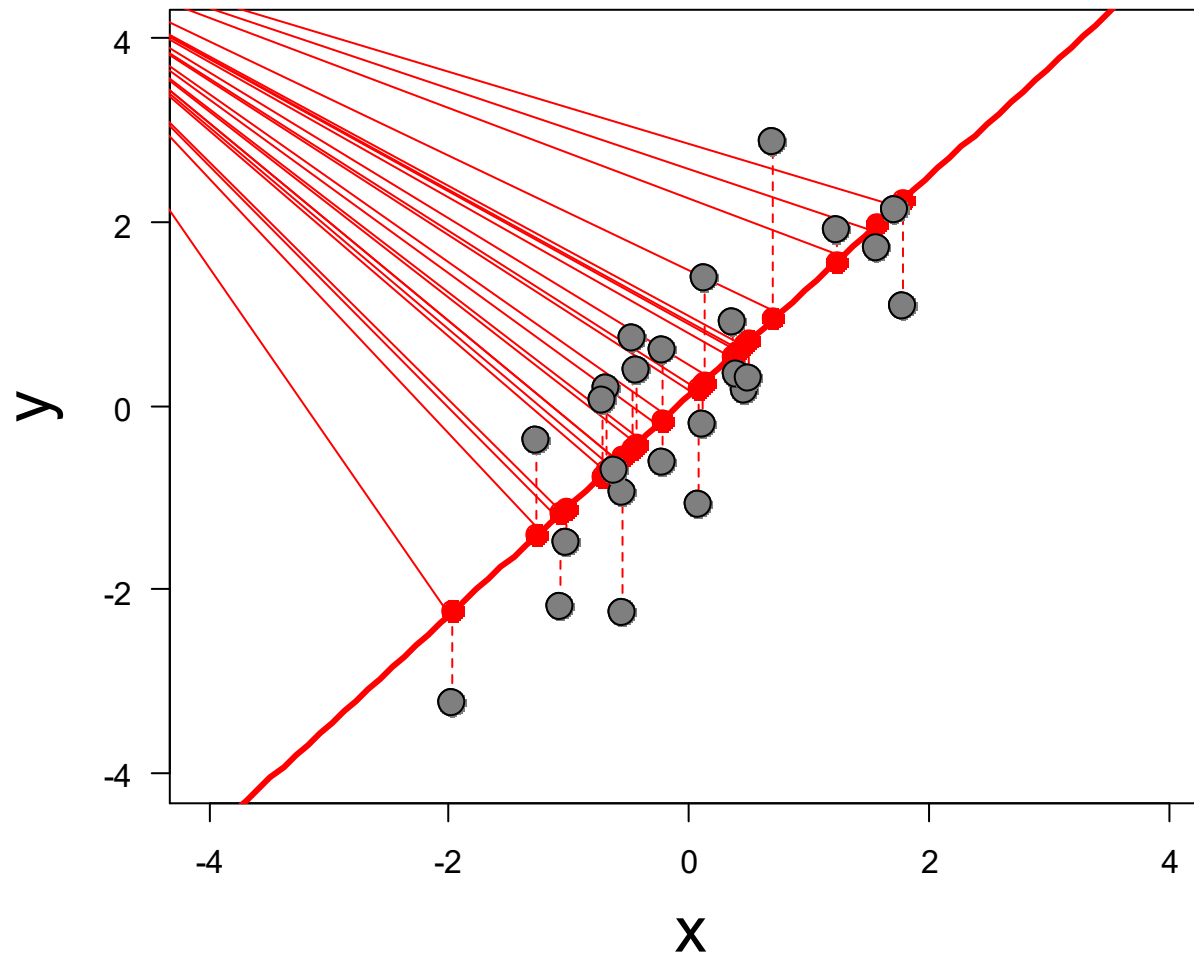
Residuals

$$\mathbf{y} - \boldsymbol{\eta}$$

Expected value of y

$$l(y_i|\boldsymbol{\beta}) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\left(\sum_{i=1}^n (y_i - \mu_i)^2\right)$$

Residual sum of squares (RSS)



Residual sum of squares (RSS)

$$\text{RSS} = \sum_{i=1}^n (y_i - \eta_i)^2$$

Add up the squared differences between data (y) and prediction (η) for every data point

As RSS increases, explanatory power declines...

AIC

AIC (general equation)

This is the log-likelihood (we work on the log-scale to prevent extreme values)

$$AIC = 2k - 2 * \ln(\hat{L})$$

k is the number of parameters (this penalizes complex models)

AICc (the lil' c means corrected for small sample sizes)

AICc

$$\text{AICc} = \text{AIC} + \frac{2k^2 + 2k}{n - k - 1}$$

This is an extra penalty for complex models with small sample sizes

If $n = 10$ and $k = 2$, then this adds up to 1.714 (a substantial penalty on the AICc scale)

If $n = 1000$ and $k = 2$, then this adds up to 0.012 (a tiny penalty on the AICc scale)

If $n = 10000$ and $k = 2$, then this adds up to 0.0012 (a teeny, weeny tiny penalty on the AICc scale)

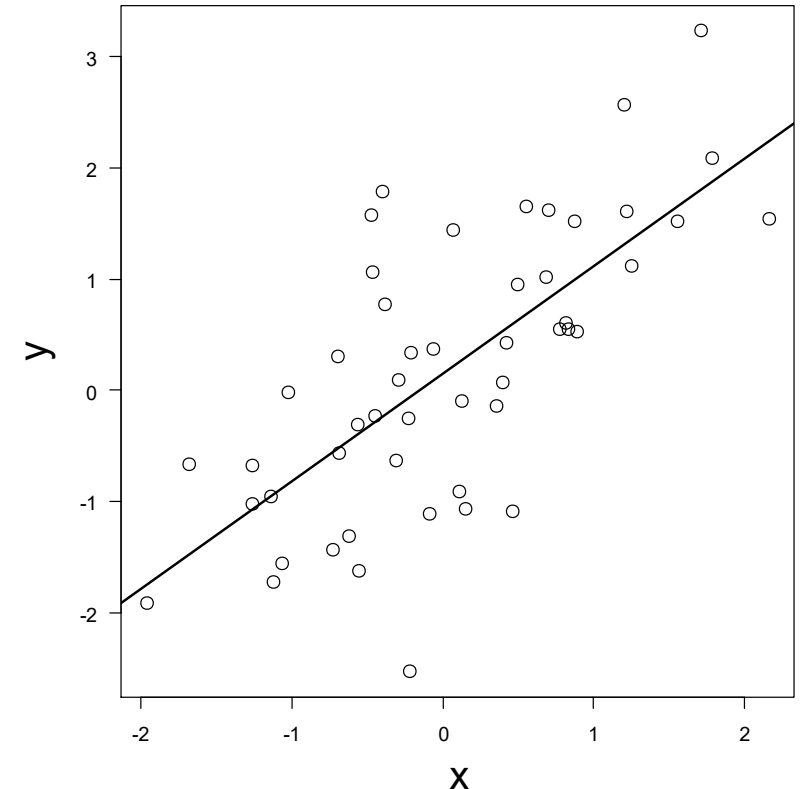
What will happen to the log-likelihood if we add a covariate that explains no information of any kind ($\beta = 0$)?

$$l(y_i | \beta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu_i)^2 \right)$$

AIC (general equation)

$$AIC = 2k - 2 * \ln(\hat{L})$$

What will happen to AIC if we add a covariate that explains no information of any kind ($\beta = 0$)?



1a. How does model selection work?

Uninformative Parameters and Model Selection Using Akaike's Information Criterion

TODD W. ARNOLD,¹ *Department of Fisheries, Wildlife and Conservation Biology, University of Minnesota, St. Paul, MN 55108, USA*

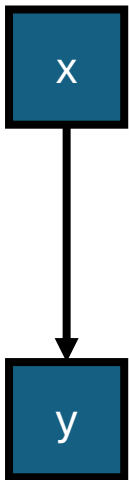
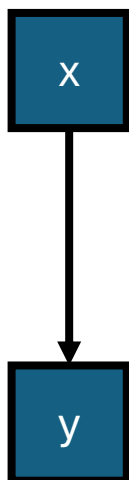


Table 1. Models examining effects of various covariates on detection probabilities of indicated breeding pairs of waterfowl in North Dakota, USA (from Pagano and Arnold 2009). I added single parameters assuming an additive effect to the base model, which included $K = 16$ parameters (8 species $\times$ 2 observers). Three of these covariates were considered biologically feasible (total ducks, vegetative cover, and cover type), 6 were not (random 5, 4, 8, and 1; not Sunday, Monday, or Wednesday; and last duck seen a mallard), and I excluded 6 additional nonsense or random variables ($\Delta AIC = 0.64\text{--}2.00$) from presentation. I evaluated all models compared to the base model using Akaike's Information Criterion (AIC), ΔAIC , and changes in model deviance (Dev).

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Simulate some data

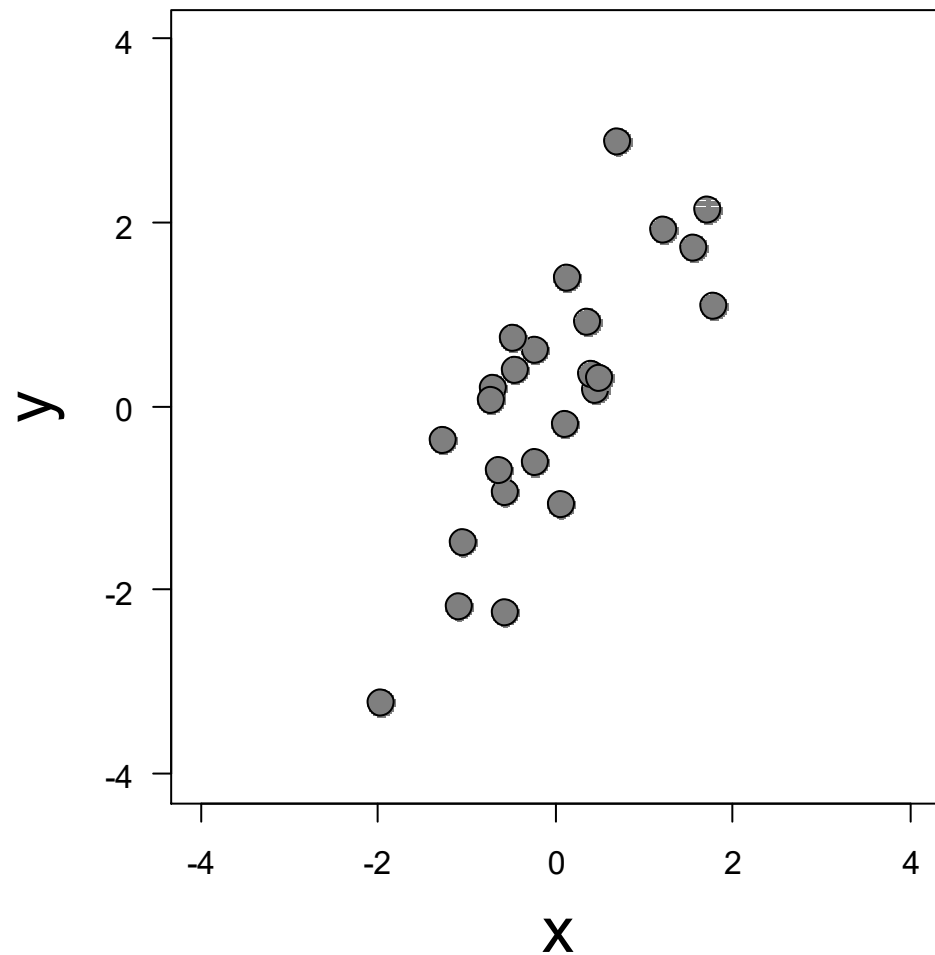


$$\boldsymbol{x} \sim \text{Normal}(0, 1)$$

$$\boldsymbol{y} \sim \text{Normal}(\boldsymbol{x}, 1)$$

$$n = 25$$

Let's do that once...



$$x \sim \text{Normal}(0, 1)$$

$$y \sim \text{Normal}(x, 1)$$

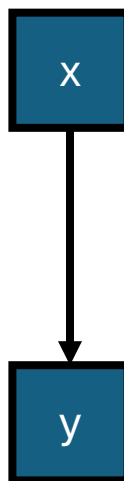
$$n = 25$$

And compare two models

m0

$$y \sim \text{Normal}(\beta_0, \sigma^2)$$

`glm(y ~ 1)`

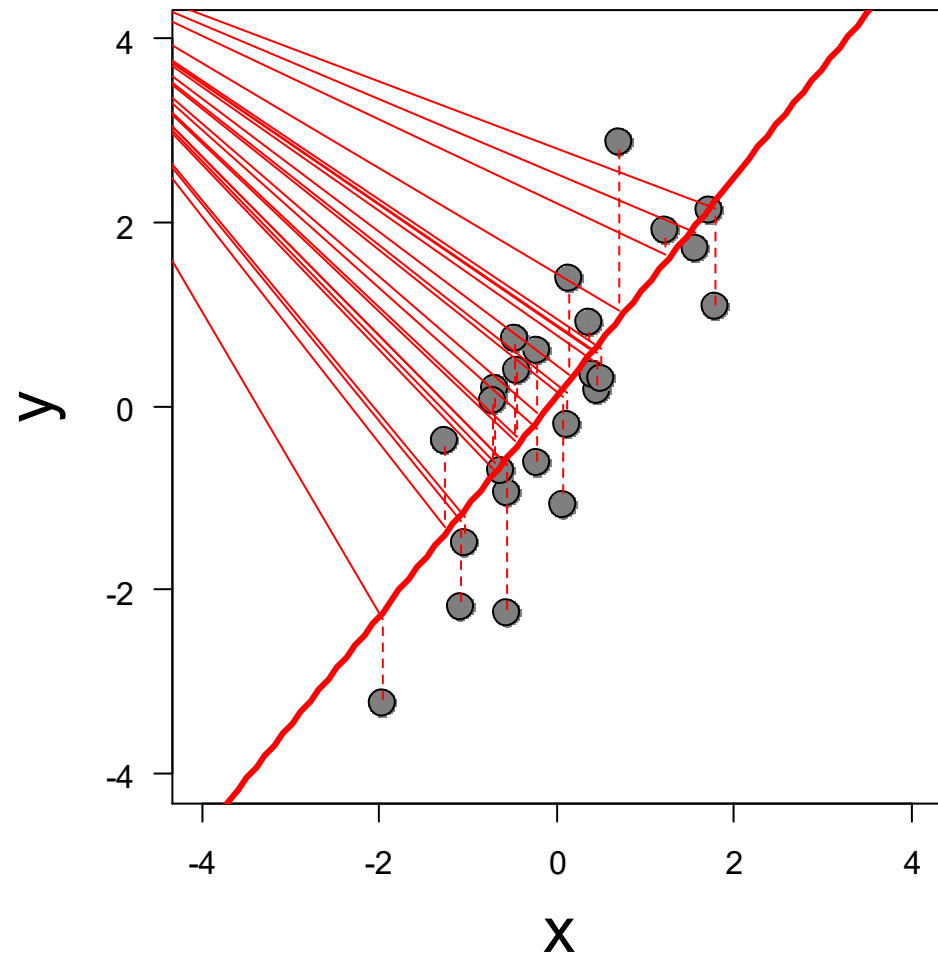
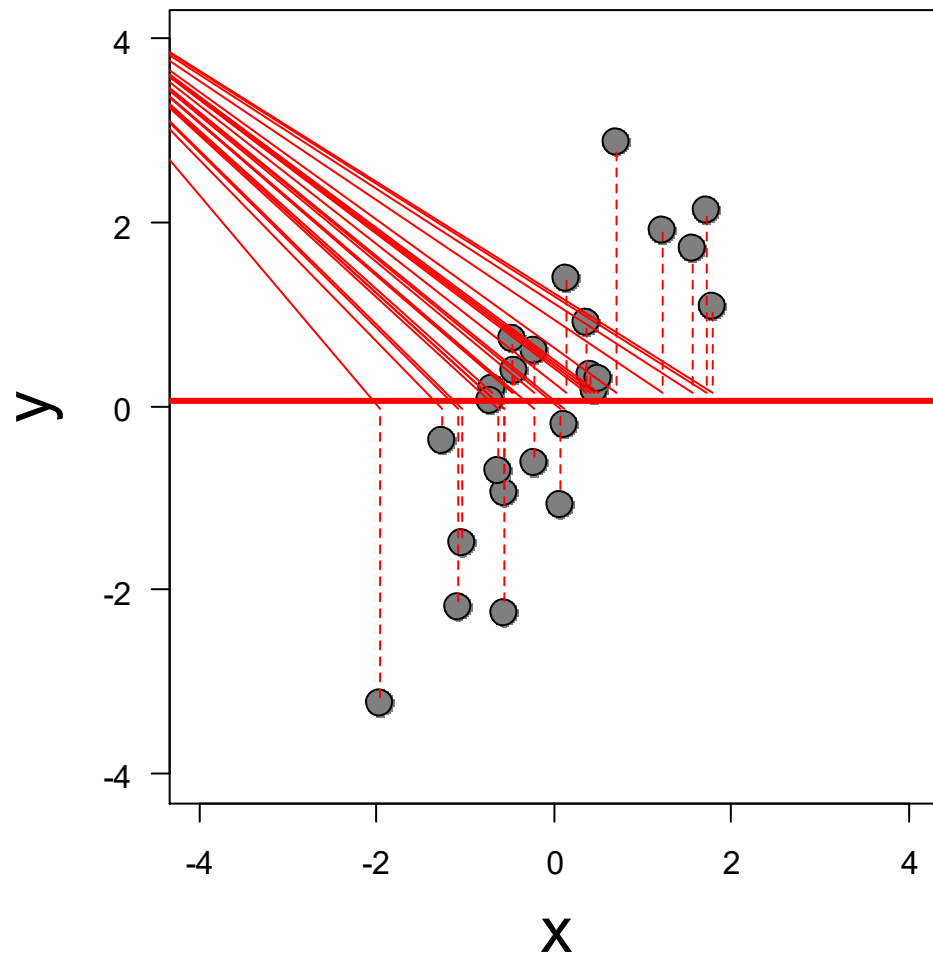


m1

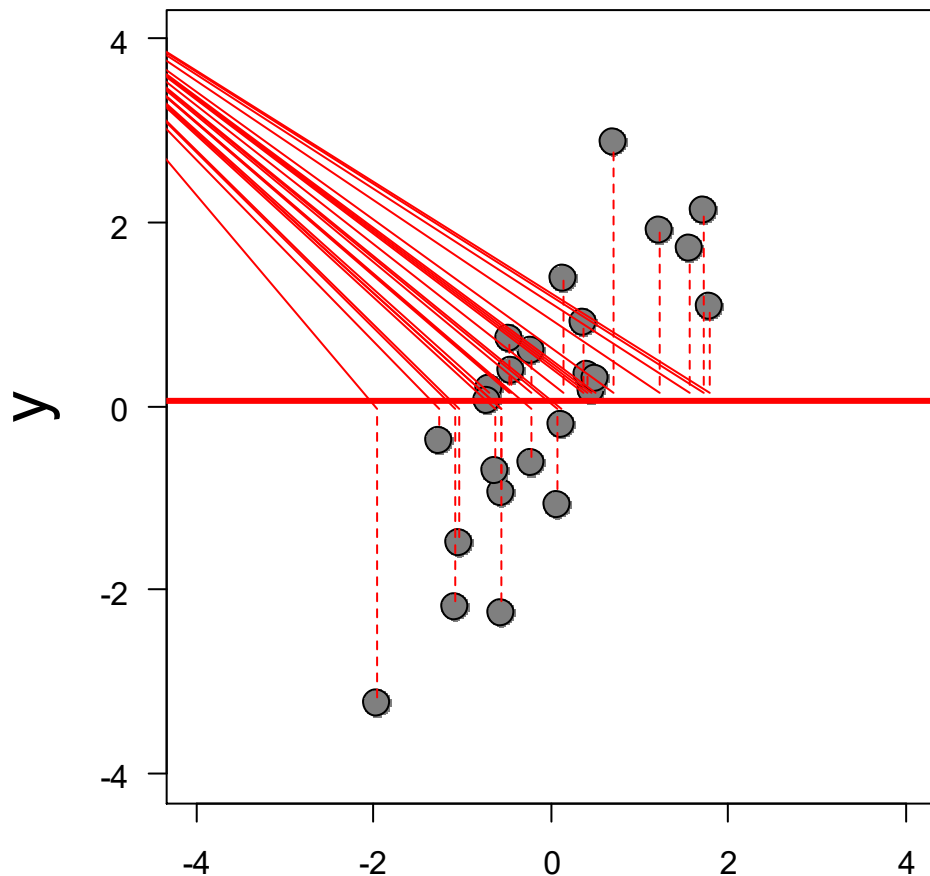
$$y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma^2)$$

`glm(y ~ x)`

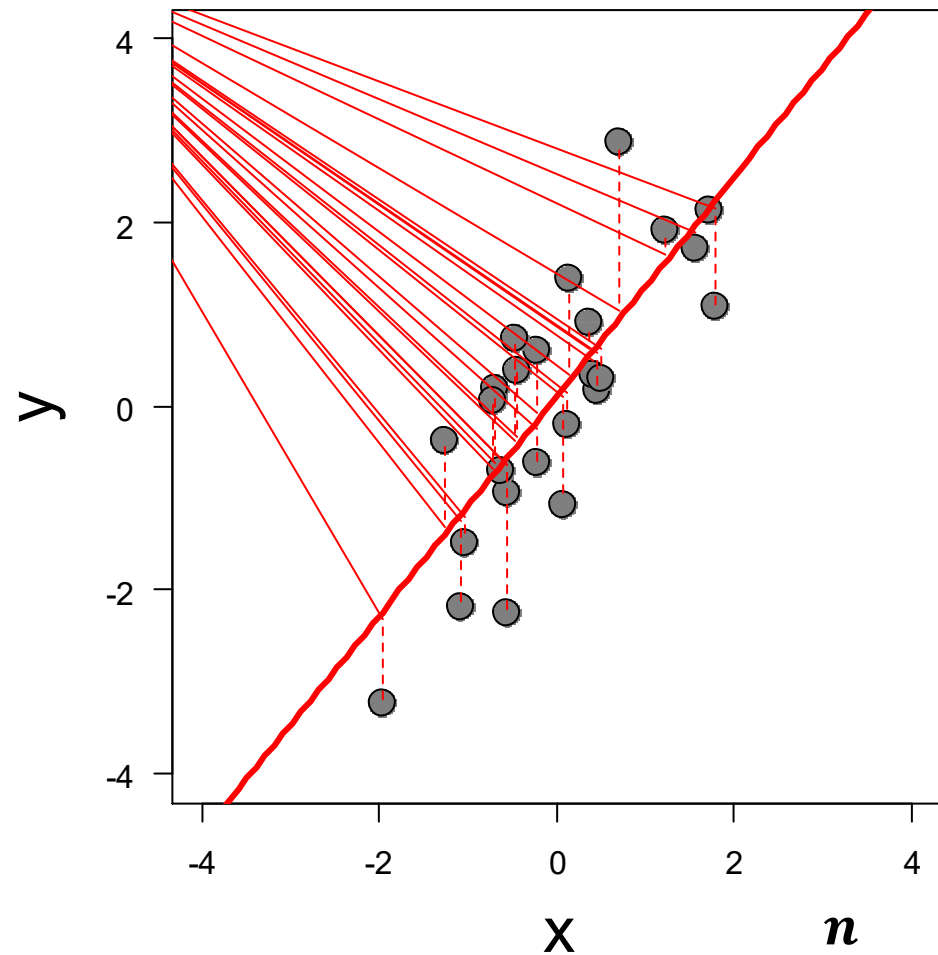
And compare two models from a single run



And compare two models from a single run (residuals)

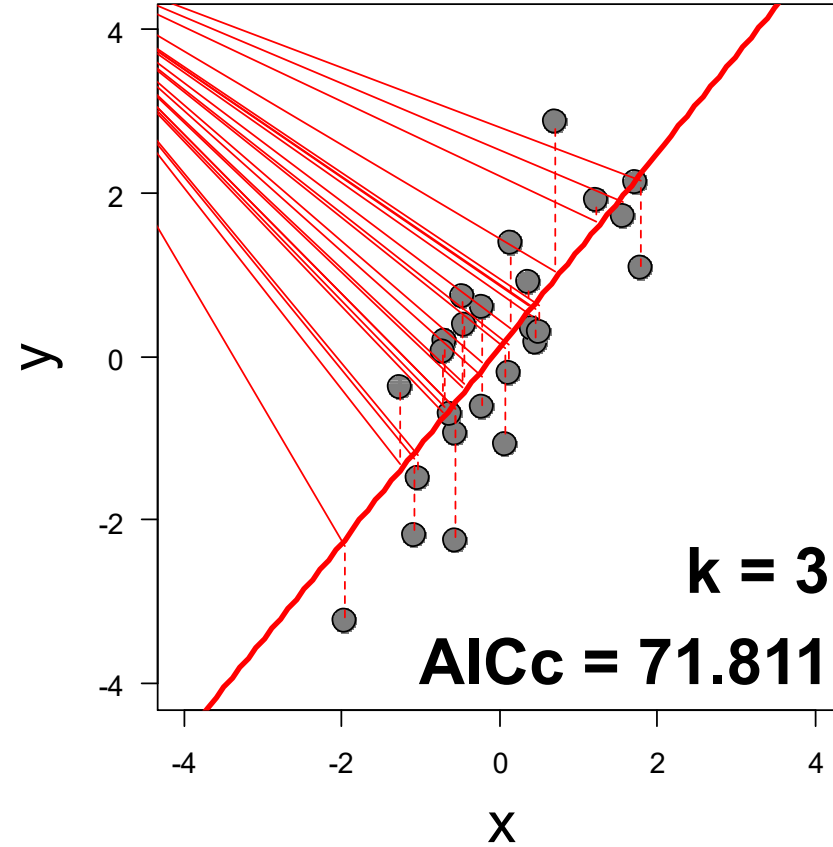
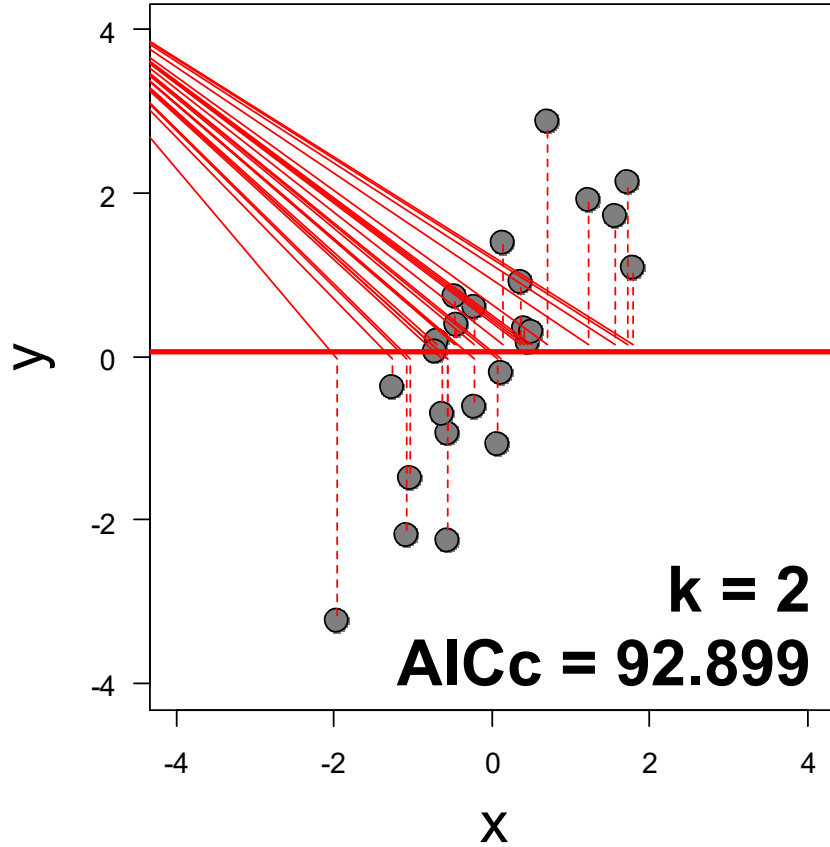


$$50.156 = \sum_{i=1}^n (y_i - \eta_i)^2$$



$$19.448 = \sum_{i=1}^n (y_i - \eta_i)^2$$

And compare two models from a single run (AICc)

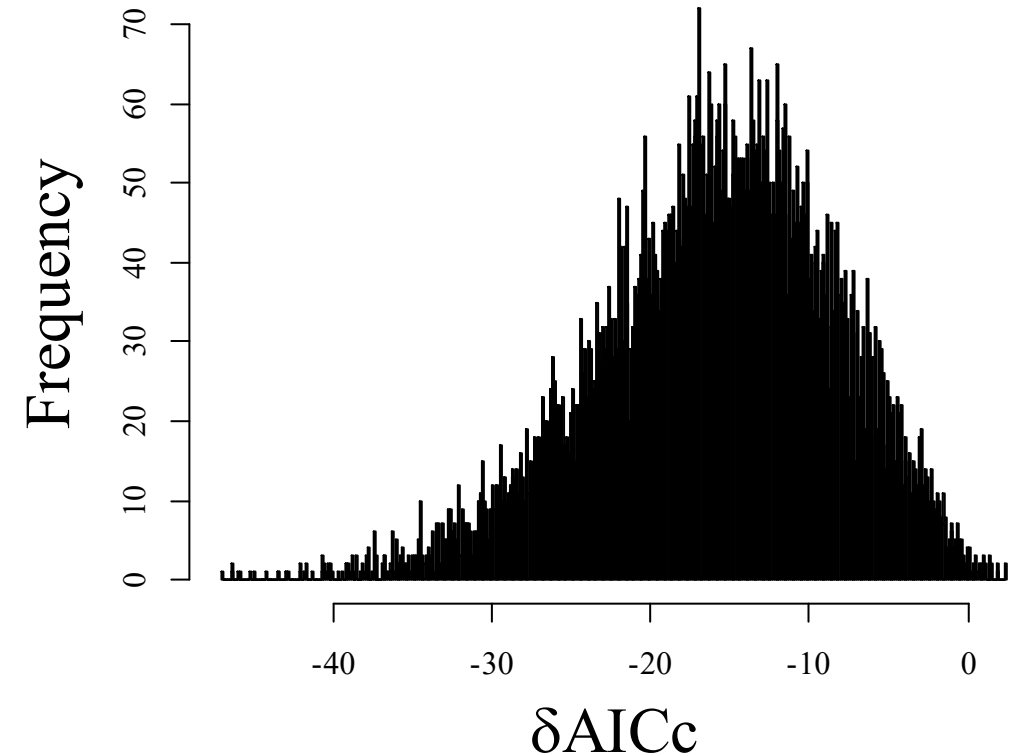


$$AICc = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{RSS}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

Cool! Let's do that 10,000 times!

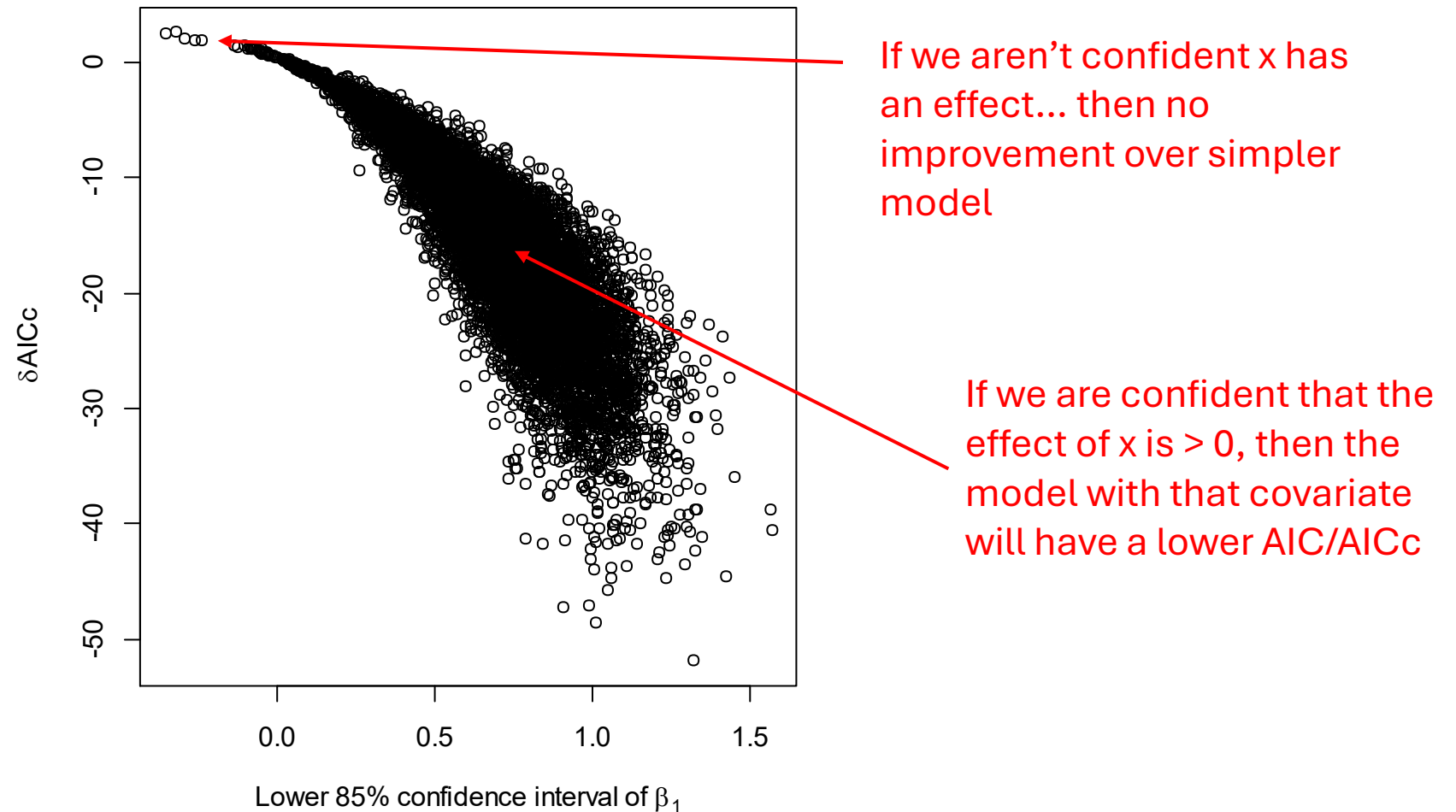
And compare two models from 10k runs

- $\text{mean}(\delta\text{AIC}) = -15.89$
- $\text{median}(\delta\text{AIC}) = -15.38$
- In 0.3% of simulations, m0 (the intercept only model) was superior to m1 (the true model).
- In 1.5% of simulations, m0 was ~equivalent to m1 ($\delta\text{AIC} < 2$).
- m0 had some support ($\delta\text{AIC} < 7$) in 11% of simulations.



The data generating model is consistently the ‘best’ model. Why not always?

Relationship between 85% confidence intervals and AIC/AICc



$$AICc = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{RSS}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

So, what are our take-homes from 'example 1a'

- **Including a parameter that has a significant (85% CI) association with the response will lead to lower AIC values (because more variance is explained).**
 - **Adding a parameter that matters will improve AIC | AICc**
- **Sometimes, it doesn't... why?**
- **In this case, it was causal. That isn't necessarily true!**

1b. How does model selection work?

let's break some stuff!

1b. Let's add some meaningless information (*m*)



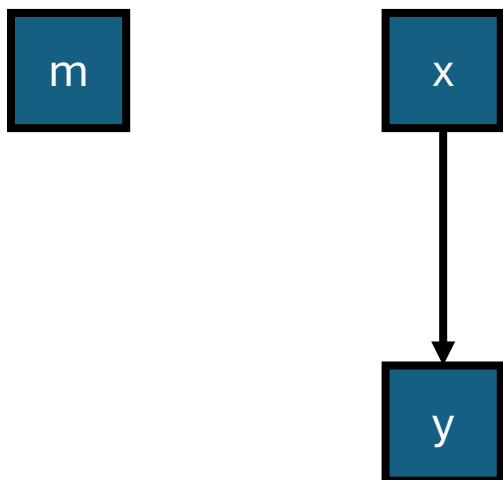
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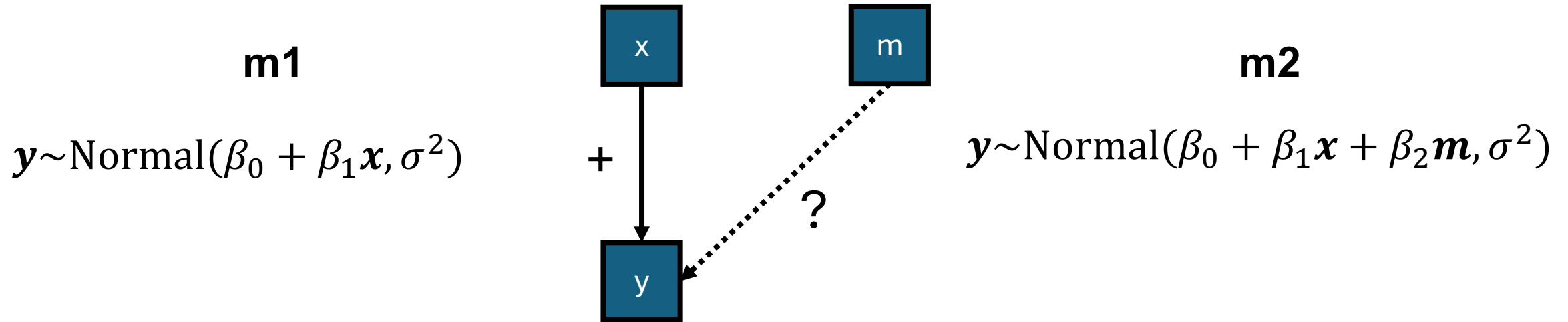
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$$n = 25$$

$$\boldsymbol{m} \sim \text{Normal}(0, 1)$$

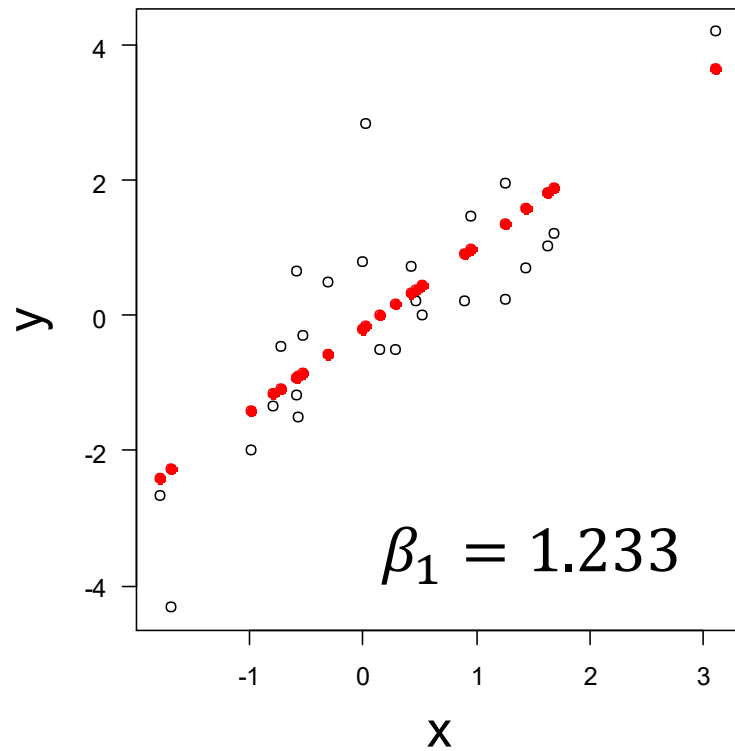
1b. Similarly, we'll compare two models*



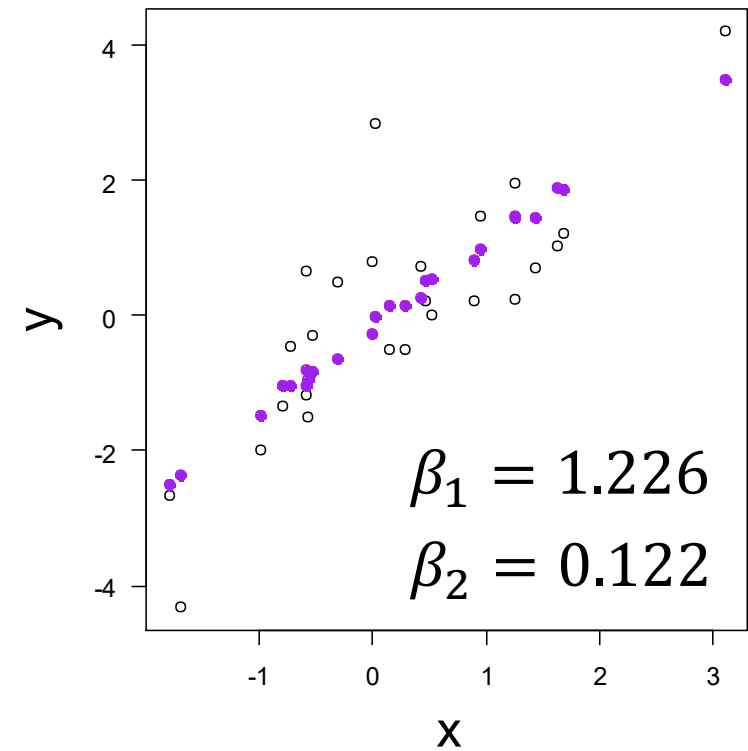
***Note that $y \sim \text{Normal}(\beta_0, \sigma^2)$ generally won't be competitive as in 1a**

1b. Predictions from both models

$$y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma^2)$$

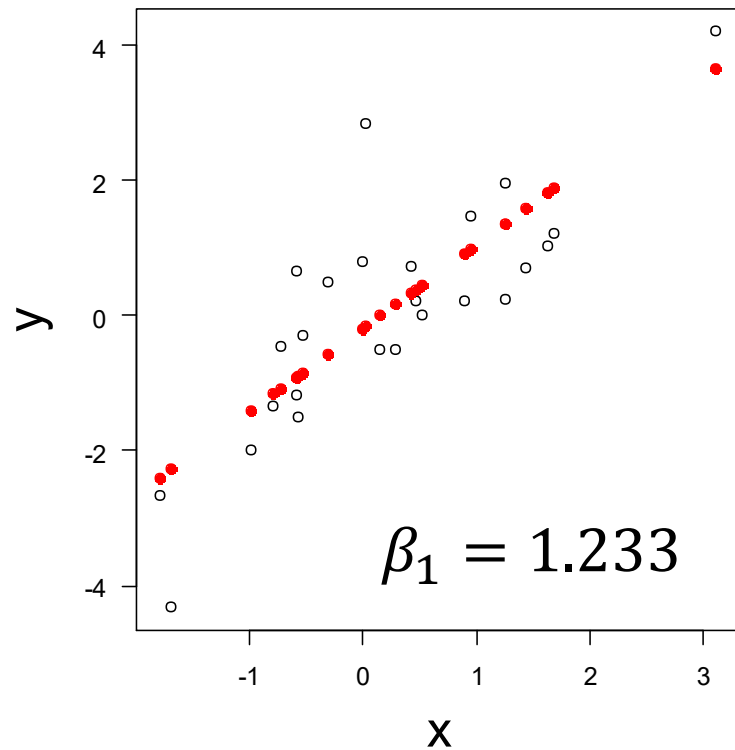


$$y \sim \text{Normal}(\beta_0 + \beta_1 x + \beta_2 m, \sigma^2)$$



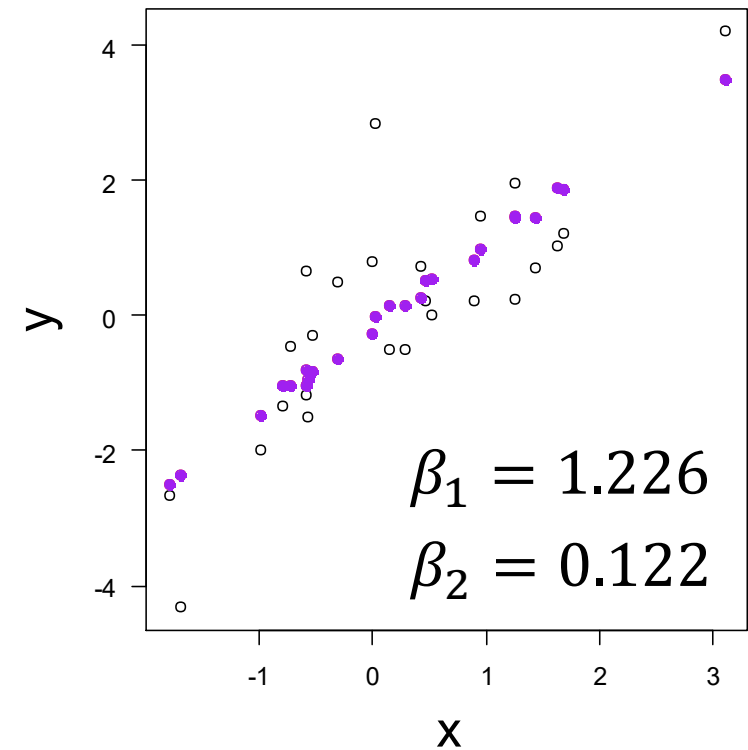
1b. RSS [purple points are a tiny bit closer]

$$y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma^2)$$



$$24.839 = \sum_{i=1}^n (y_i - \eta_i)^2$$

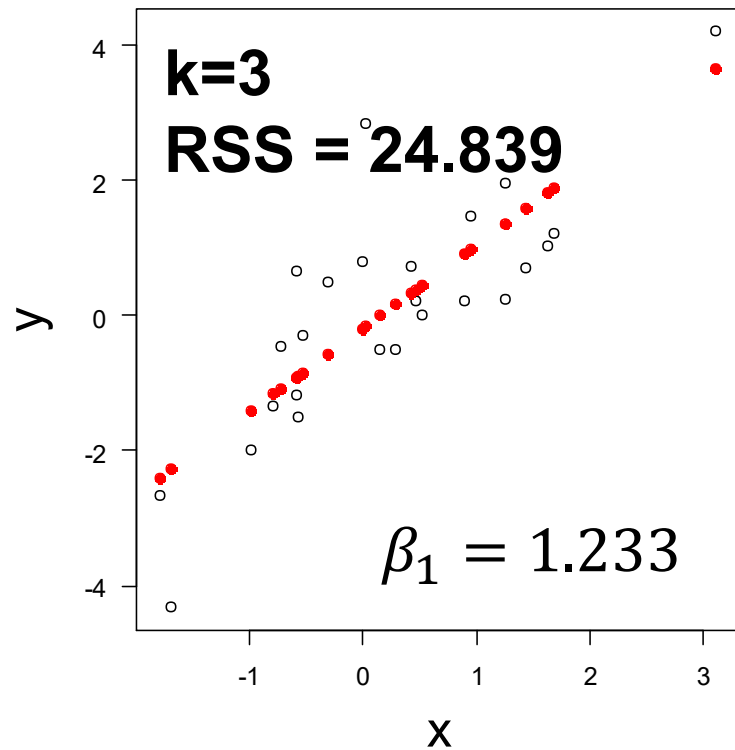
$$y \sim \text{Normal}(\beta_0 + \beta_1 x + \beta_2 m, \sigma^2)$$



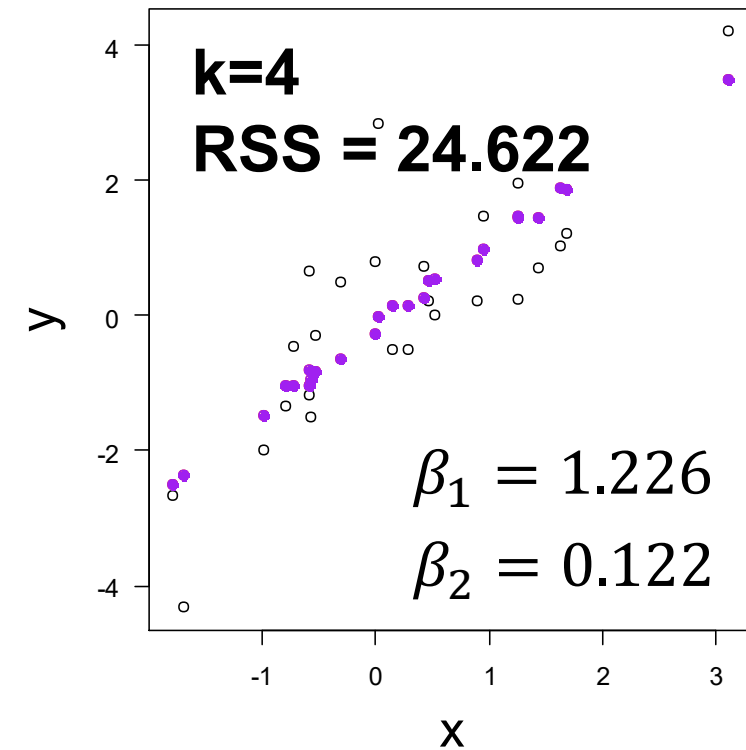
$$24.622 = \sum_{i=1}^n (y_i - \eta_i)^2$$

1b. What are our predictions for AIC?

$$y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma^2)$$



$$y \sim \text{Normal}(\beta_0 + \beta_1 x + \beta_2 m, \sigma^2)$$

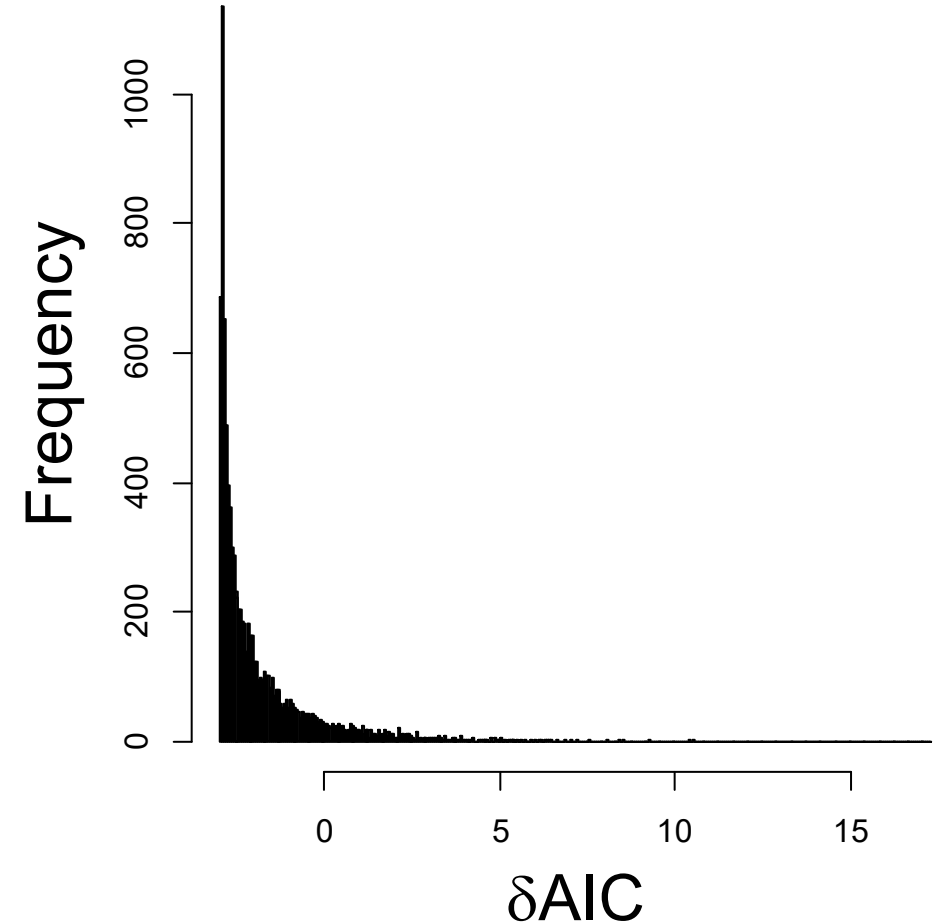


$$\text{AICc} = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{\text{RSS}}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

Here are results from 10k simulations

- $\text{mean}(\delta\text{AIC}) = -1.704$
- $\text{median}(\delta\text{AIC}) = -2.316$
- In 11.4% of simulations, m2 (the two covariate model) was superior to m1 (the true model).
- In 38.5% of simulations, m2 was ~equivalent to m1 ($\delta\text{AIC} < 2$).
- **m2 had some support ($\delta\text{AIC} < 4$) in 100% of simulations.**

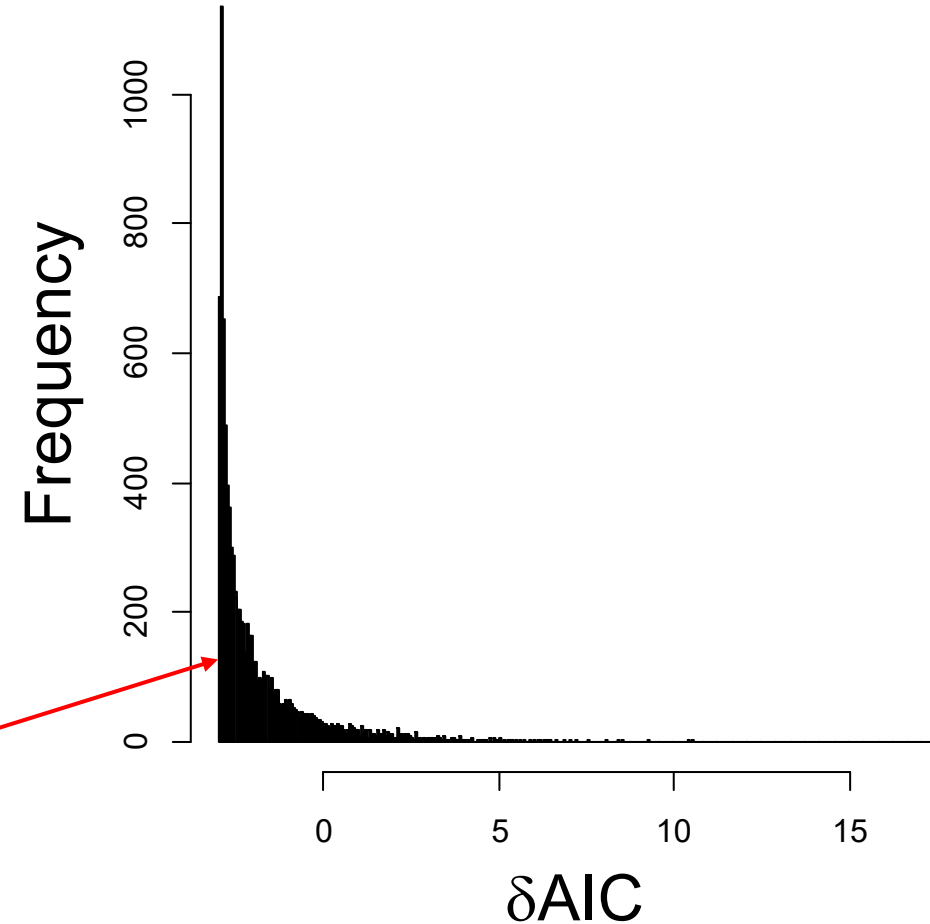
Did it?!



What's happening here?

- m2 had some support ($\delta\text{AIC} < 4$) in 100% of simulations.
- Did it?!

Cual es el problema?



$$\text{AICc} = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{\text{RSS}}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

What factors affect m2 (model including m) being 'important'?

$$y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma^2)$$

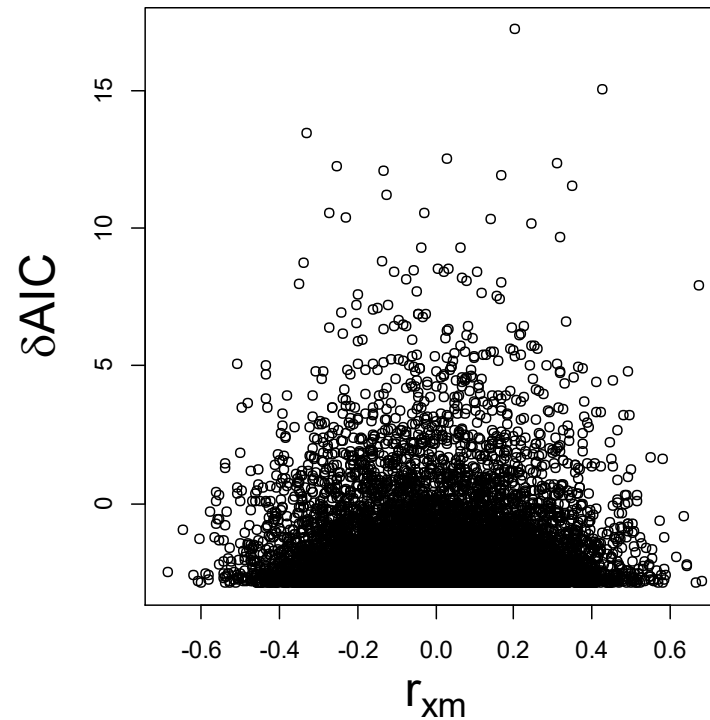
$$y \sim \text{Normal}(\beta_0 + \beta_1 x + \beta_2 \mathbf{m}, \sigma^2)$$

$$\text{AICc} = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{\text{RSS}}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

What factors affect m2 (model including m) being 'important'?

$$y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma^2)$$

$$y \sim \text{Normal}(\beta_0 + \beta_1 x + \beta_2 m, \sigma^2)$$

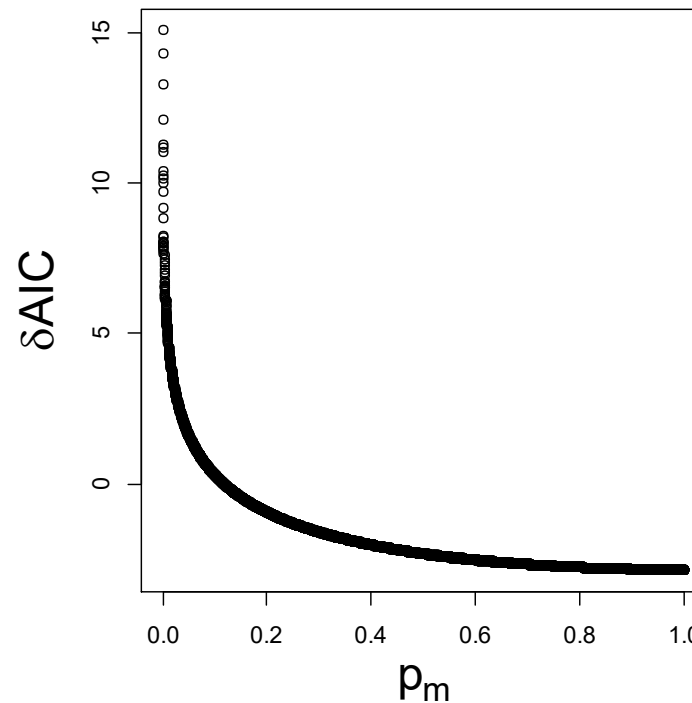


$$AICc = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{RSS}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

What factors affect m2 (model including m) being 'important'?

$$y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma^2)$$

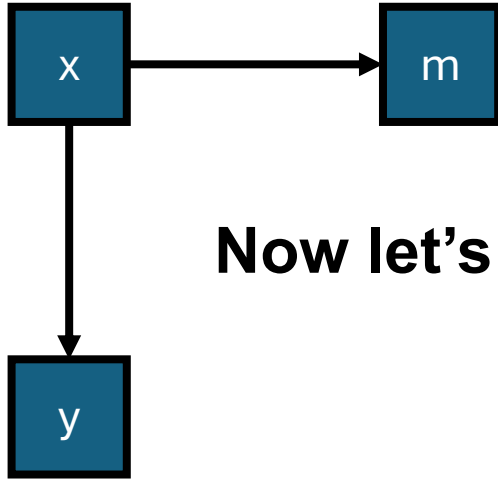
$$y \sim \text{Normal}(\beta_0 + \beta_1 x + \beta_2 m, \sigma^2)$$



The effect of m had a p-value < 0.1 in approximately 10% of simulations (that makes sense!)

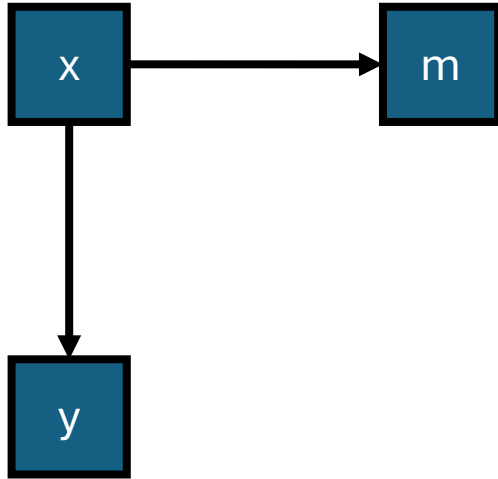
δAIC favored the two covariate model 11.4% of the time...

$$\text{AICc} = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{\text{RSS}}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$



Now let's really break some stuff (10k times)!

How do complex models affect model selection? A simple simulated example



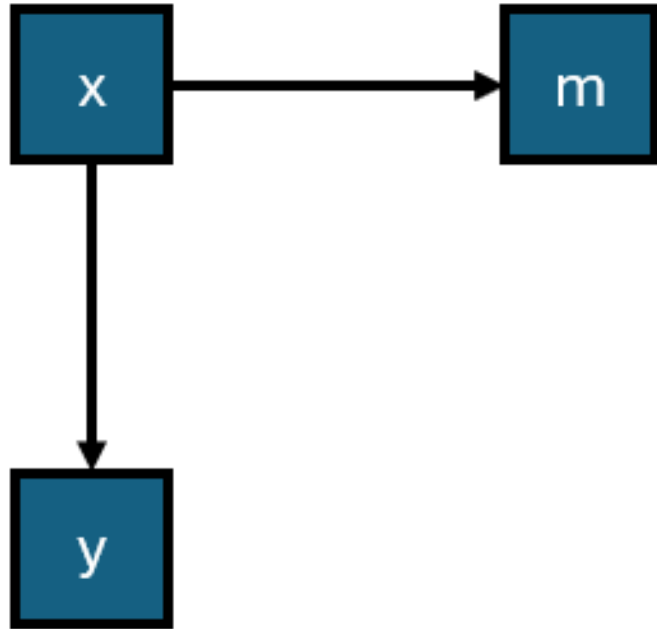
$$\boldsymbol{x} \sim \text{Normal}(0, 1)$$

$$\boldsymbol{y} \sim \text{Normal}(\boldsymbol{x}, 1)$$

$$\boldsymbol{m} \sim \text{Normal}(\boldsymbol{x}, 1)$$

$$n = 25$$

Now we'll imagine that we misunderstand the data-generating process (fun!)



$$\boldsymbol{x} \sim \text{Normal}(0, 1)$$

$$\boldsymbol{y} \sim \text{Normal}(\boldsymbol{x}, 1)$$

$$\boldsymbol{m} \sim \text{Normal}(\boldsymbol{x}, 1)$$

$$n = 25$$

Let's assume that we think ***m*** affects ***y***

This isn't as wild as it seems

Eiser et al. (1979) *Addictive Behavior*

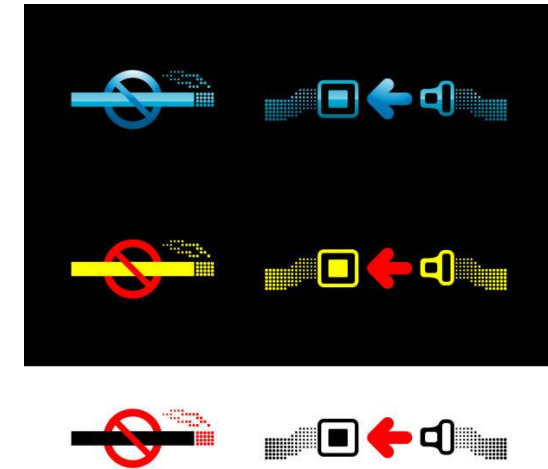
Grout et al. (1983) *Public Health*

Cliff et al. (1982) *Public Health*

Helsing and Comstock (1977) *American Journal of Public Health*

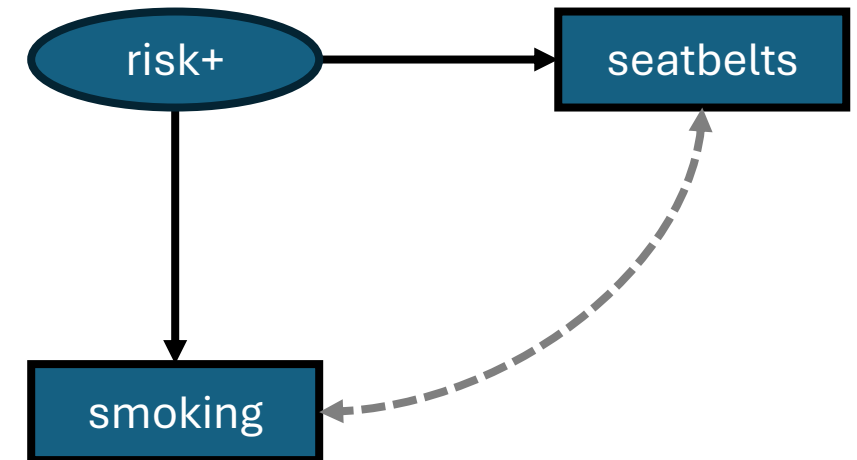
Manheimer et al. (1966) *Traffic Safety Research Review*

Williams (1973) *Journal of Health and Social Behavior*



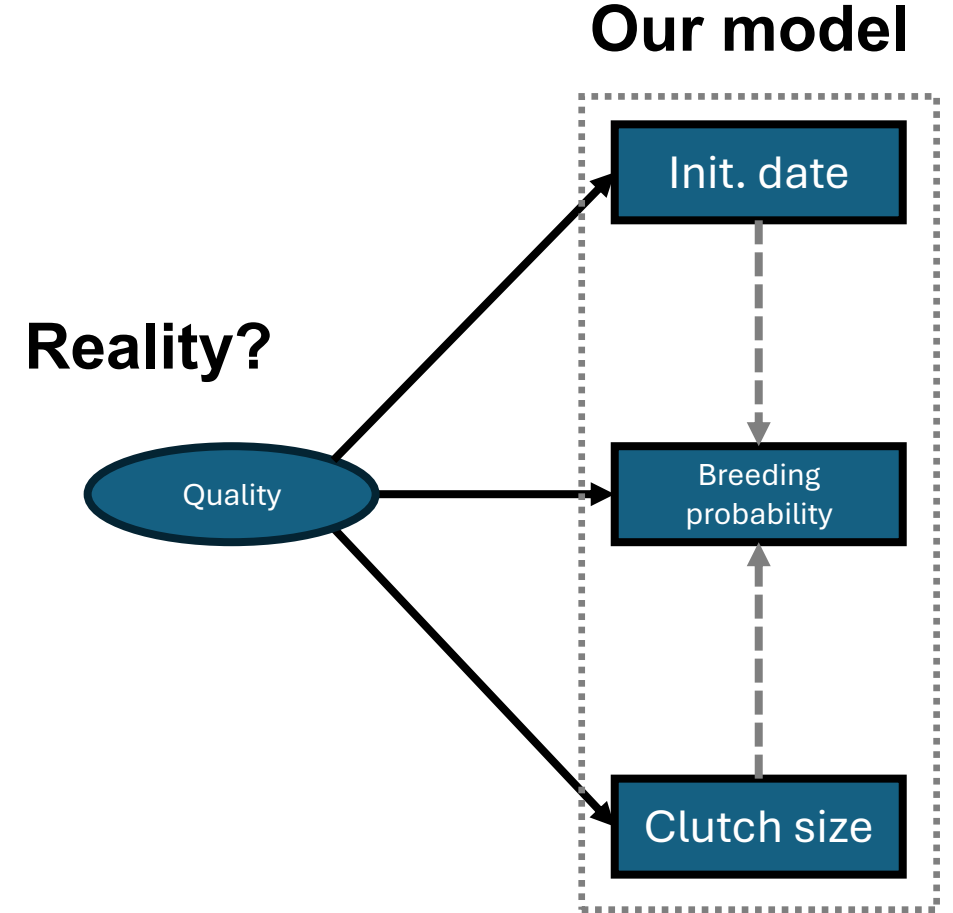
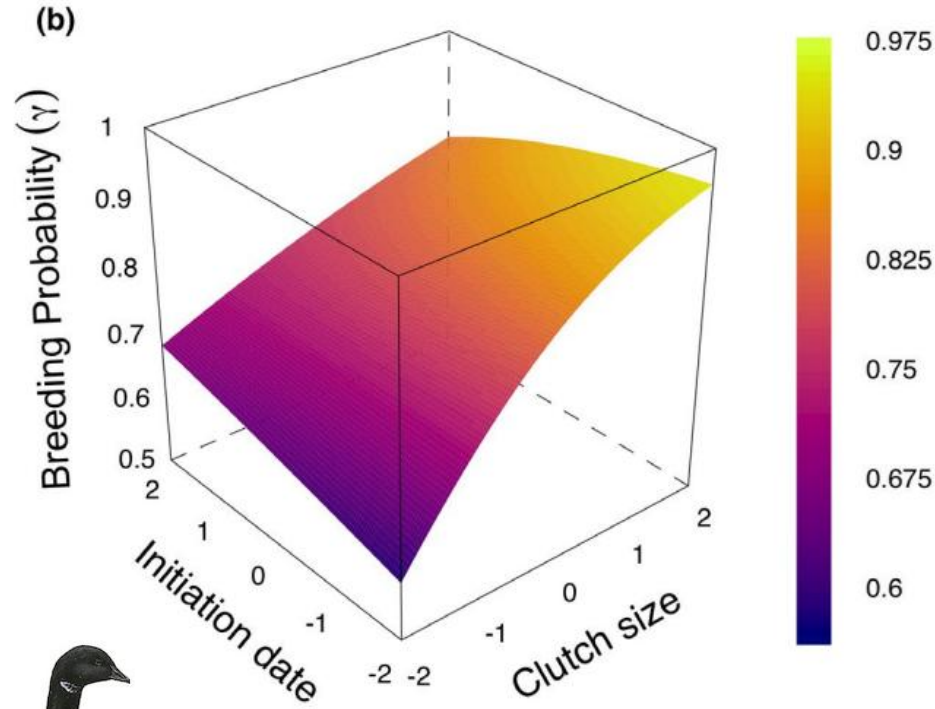
Indeed, in a study conducted in 2006 as part of a tobacco litigation, seat-belt usage was listed as one of the first variables to be controlled for.

-Pearl and Mackenzie (2018) *The Book of Why*



Smoking and seat-belt use

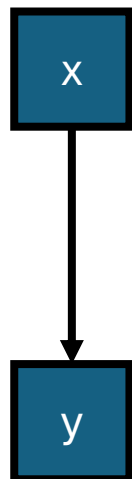
This isn't as wild as it seems



Lohman, Riecke, et al. (2021) *Ecology and Evolution*

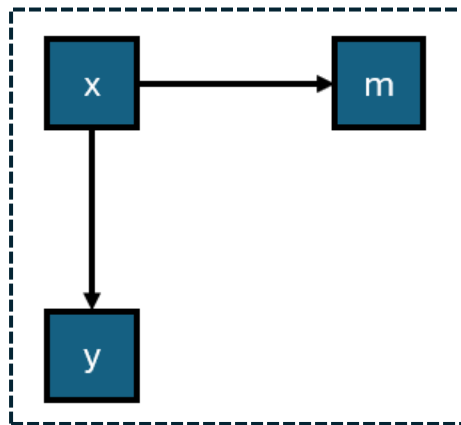
Back to our simulation, we'll compare two models (power analysis)

m1



$$y \sim \text{Normal}(\beta_0 + \beta_1 x, \sigma^2)$$

Reality



m2



$$y \sim \text{Normal}(\beta_0 + \beta_1 \mathbf{m}, \sigma^2)$$

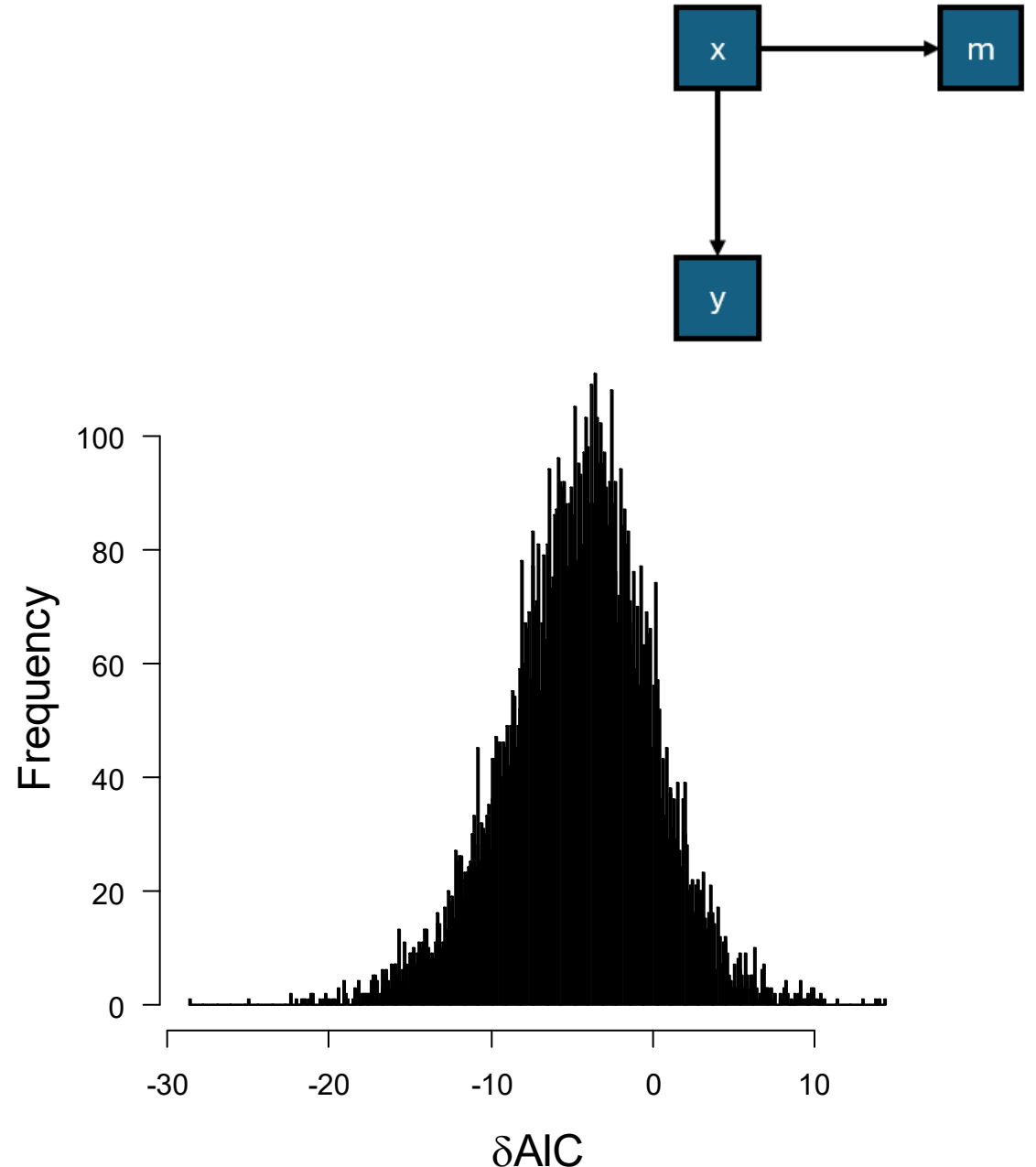
We'll do this 10k times (brute force to the rescue!) and save:

- 1. The difference in AICc between the models (δAIC)**
 - Differences of < 2 indicate model 'equivalence'
 - Differences of < 7 indicate some support and should rarely be dismissed
- 2. The model weights (a function of δAIC that tells us the relative likelihood of each model)**
- 3. The observed correlation between x and m**

Burnham, Anderson, and Huyvaert (**2011**) *Behavioral Ecology and Sociobiology*

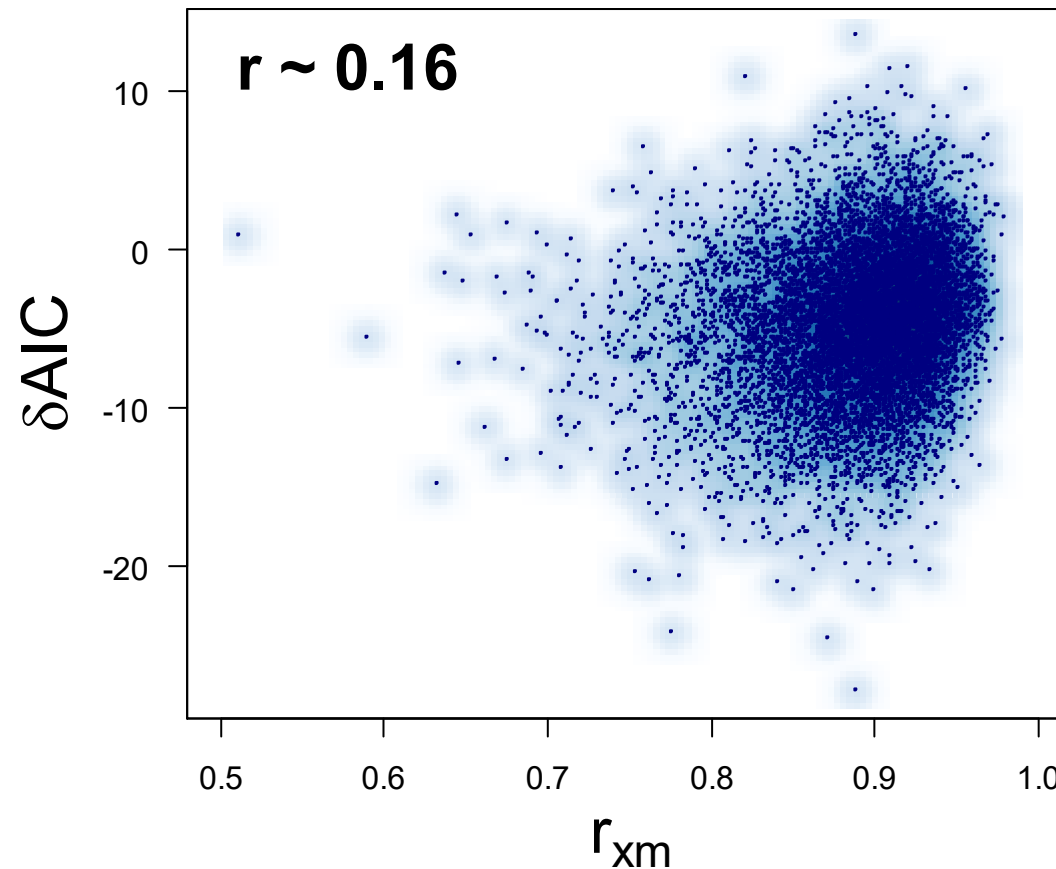
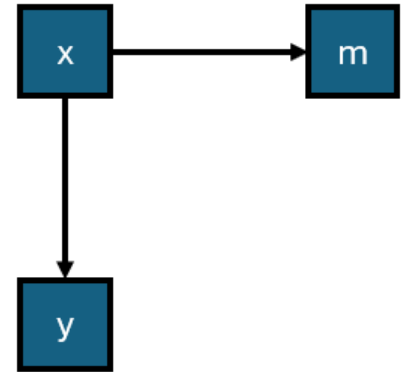
Results (n = 25; 10k sims)

- $\text{mean}(\delta\text{AIC}) = -4.57$
- $\text{median}(\delta\text{AIC}) = -4.35$
- In 14% of simulations, m2 (the spurious model) was superior to m1 (the true model).
- In 28% of simulations, m2 was ~equivalent to m1 ($\delta\text{AIC} < 2$).
- m2 had some support ($\delta\text{AIC} < 7$) in 72% of simulations.

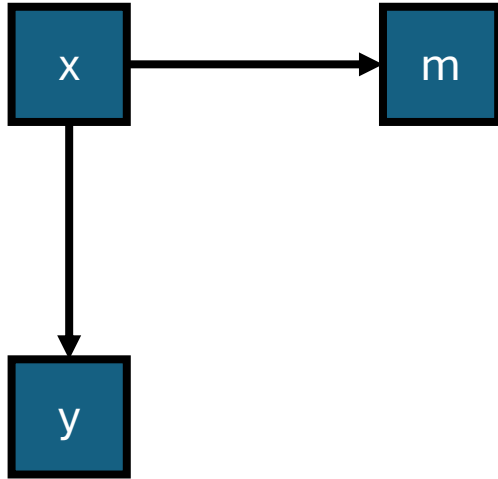


Results (n = 25; 10k sims)

- Increases in correlations among x and m led to difficulty in resolving differences among competing models



What would happen if we changed n ?



$$\boldsymbol{x} \sim \text{Normal}(0, 1)$$

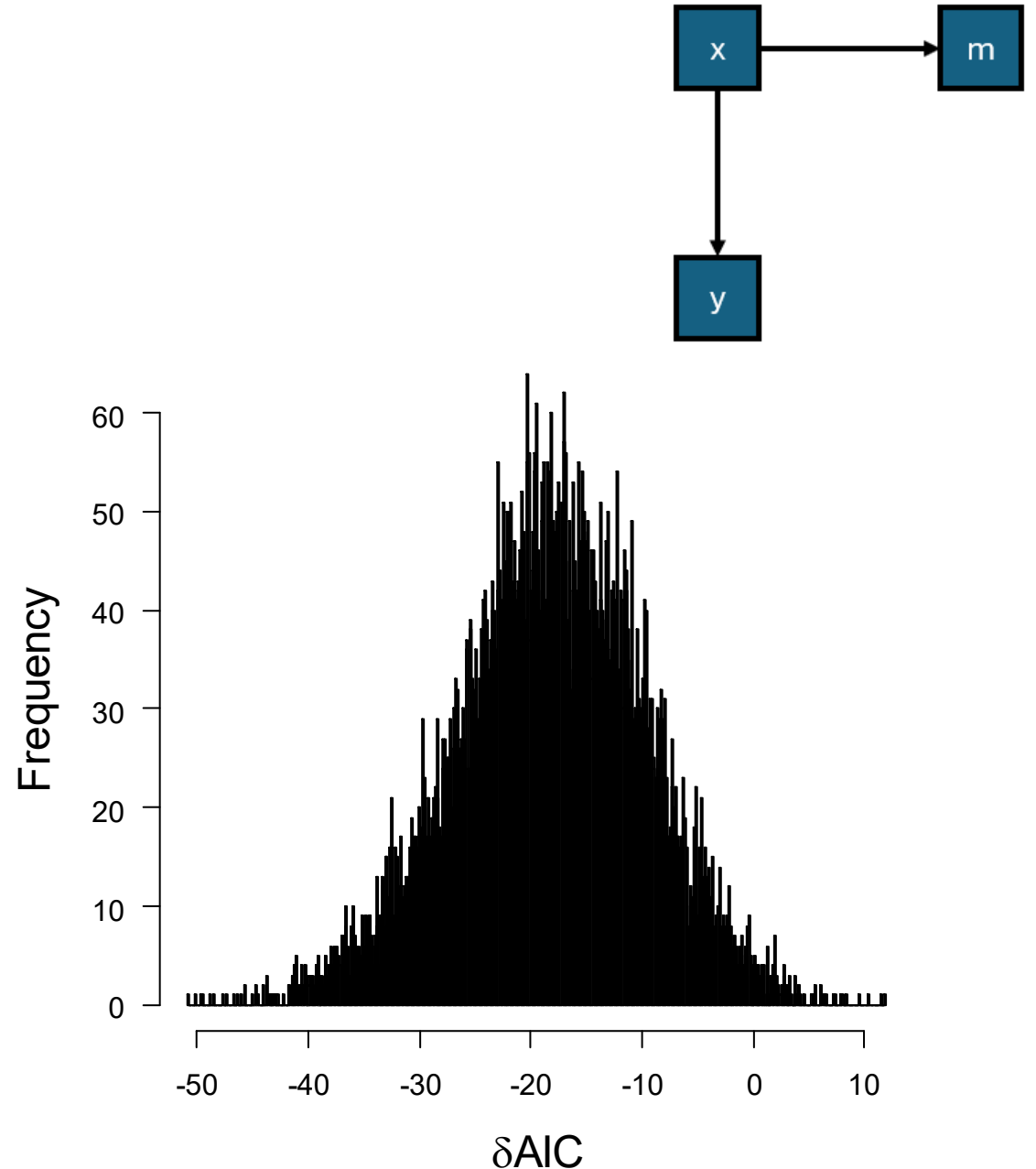
$$\boldsymbol{y} \sim \text{Normal}(\boldsymbol{x}, 1)$$

$$\boldsymbol{m} \sim \text{Normal}(\boldsymbol{x}, 1)$$

$$n = \textcolor{red}{25} \ 100$$

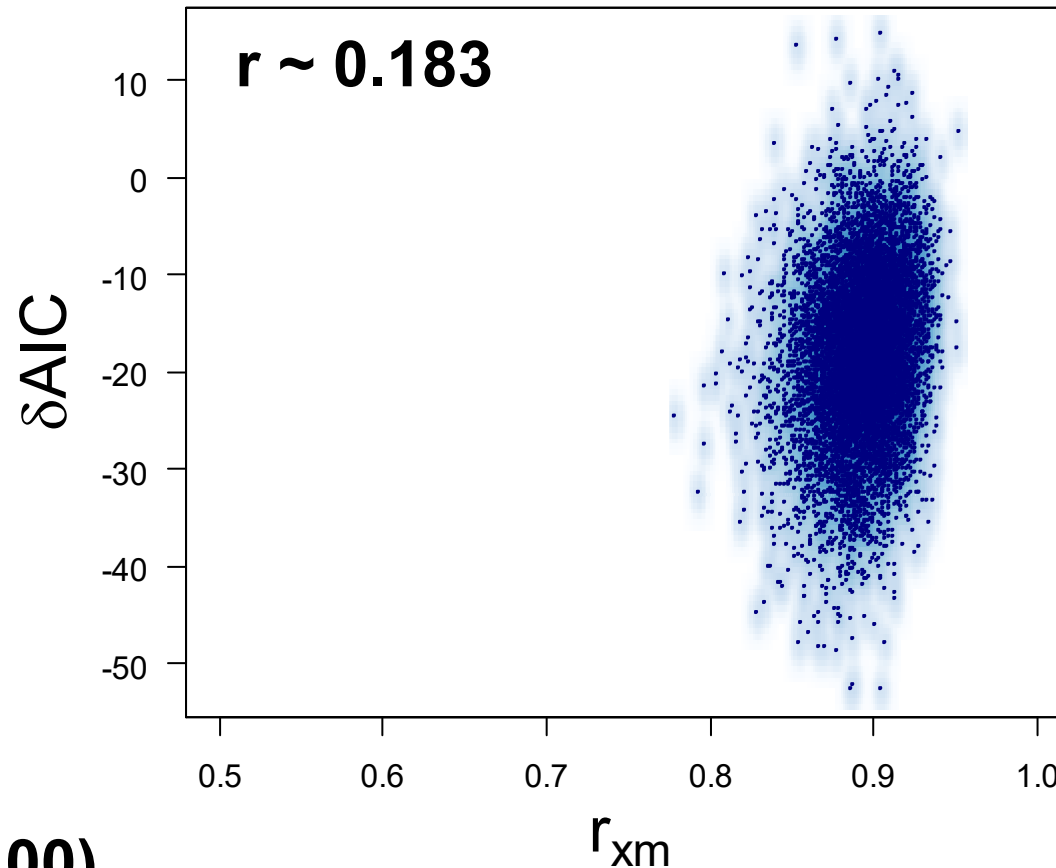
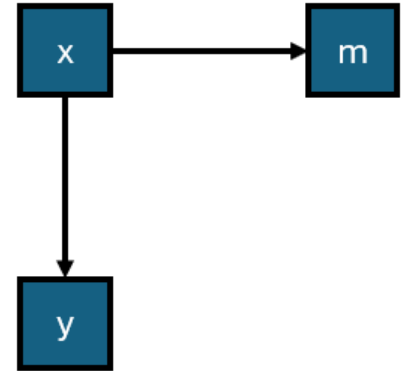
Results (n = 100; 10k sims)

- $\text{mean}(\delta\text{AIC}) = -18.28$
- $\text{median}(\delta\text{AIC}) = -18.13$
- In 1% of simulations, m2 (the spurious model) was superior to m1 (the true model).
- In 2% of simulations, m2 was equivalent to m1 ($\delta\text{AIC} < 2$).
- m2 had some support ($\delta\text{AIC} < 7$) in 8% of simulations.



Results (n = 100; 10k sims)

- Increases in correlations among x and m led to difficulty in resolving differences among competing models



It works! (with n = 100)

Ok, deep breath....

There is a clear relationship between the amount of information explained by a covariate and model performance

If a covariate explains information, the parameter estimate will be 'significant'

If the parameter estimate is significant (at 85% CIs), AIC will improve (decrease). If it doesn't, then the parameter isn't doing much...



There is nothing ‘wrong’ with AIC (or DIC, BIC, WAIC, LOOIC)*, but...

- It is hard to get the ‘right’ answer and we need large samples sizes (n)

Each technique has faced some valid criticism, nothing is ‘perfect’

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- If we’re interested in annual estimates, large n ’s are hard to obtain. Each additional year leads to $n+1$
- These are simple univariate regression models! Model selection will not save you from ‘bad ideas’ in complex systems.
- AIC/AICc (+ others) are only comparable if the same response variables are used!

***Each technique has faced some valid criticism, nothing is ‘perfect’**

Wrapping up AICc

1. *What is a 'meaningful' difference in AICc? 2, 3, 4, 7?*

AICc (for linear models)

$$AICc = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{RSS}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

Wrapping up AICc

1. *What is a 'meaningful' difference in AICc? 2, 3, 4, 7?*

- **This is somewhat dependent on n and k.**

$$AICc = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{RSS}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

Wrapping up AICc

1. *What is a 'meaningful' difference in AICc? 2, 3, 4, 7?*

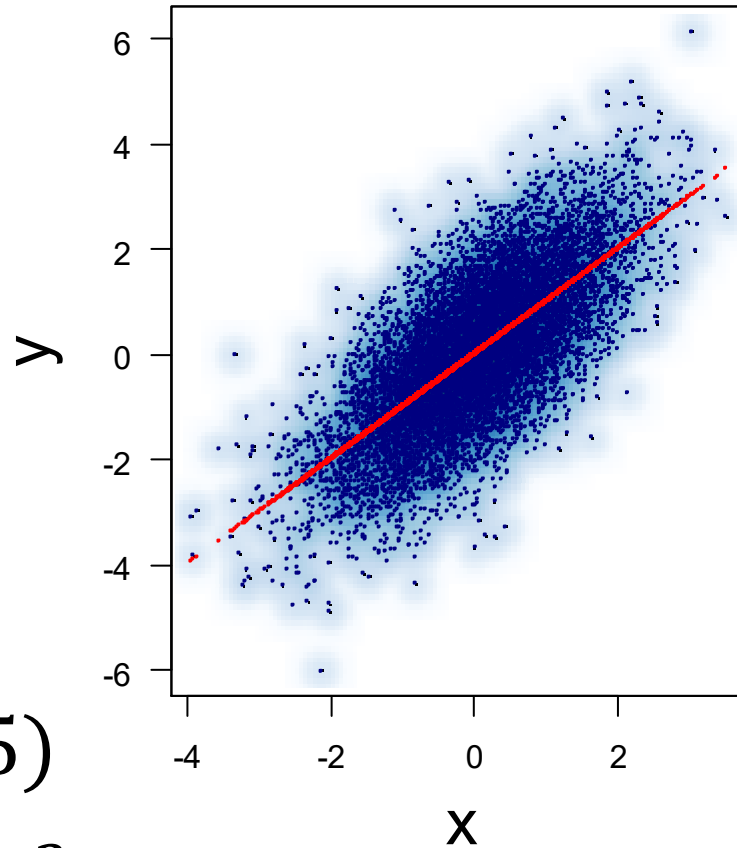
- **This is somewhat dependent on n and k.**

2. *What does a 'meaningful' difference in AICc mean?*

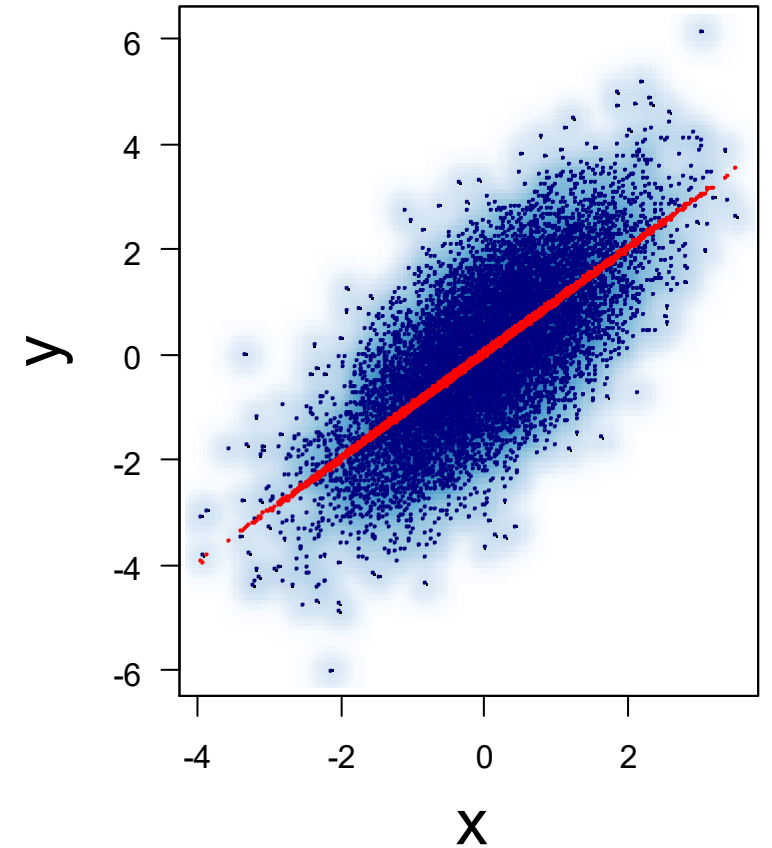
$$\text{AICc} = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{\text{RSS}}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

A final AICc vignette... what do you think δAICc is for these two models?

`lm(y ~ x)`



`lm(y ~ x + w)`



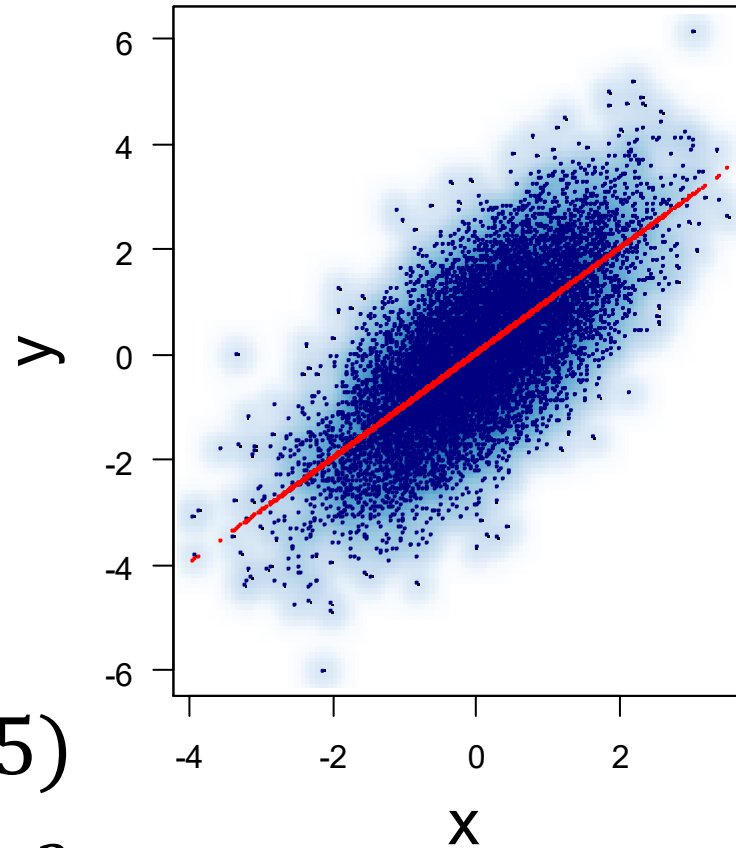
$x \sim \text{Normal}(0, 1)$

$w \sim \text{Normal}(0, 0.025)$

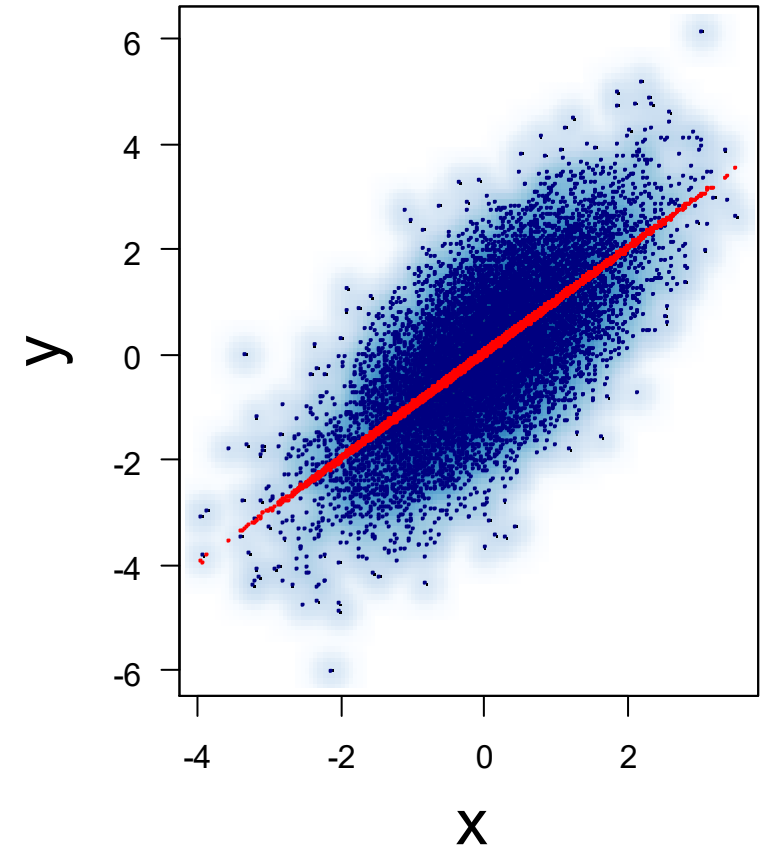
$y \sim \text{Normal}(x + w, \sigma^2 = 1)$

A final AICc vignette... what do you think δAICc is for these two models?

AICc = 28451



AICc = 28440



$x \sim \text{Normal}(0, 1)$

$w \sim \text{Normal}(0, 0.025)$

$y \sim \text{Normal}(x + w, \sigma^2 = 1)$

Wrapping up AICc

1. *What is a 'meaningful' difference in AICc? 2, 3, 4, 7?*

- **This is somewhat dependent on n and k.**

2. *What does a 'meaningful' difference in AICc mean?*

The reduction in RSS...

$$\text{AICc} = 2k + n \times \left(\ln \left(2 \times \pi \times \frac{\text{RSS}}{n} \right) + 1 \right) + \frac{2k^2 + 2k}{n - k - 1}$$

offsets the penalties for adding complexity.

Wrapping up AICc

1. *What is a 'meaningful' difference in AICc? 2, 3, 4, 7?*

- **This is somewhat dependent on n and k .**

2. *What does a 'meaningful' difference in AICc mean?*

- **This is somewhat dependent on n and k .**

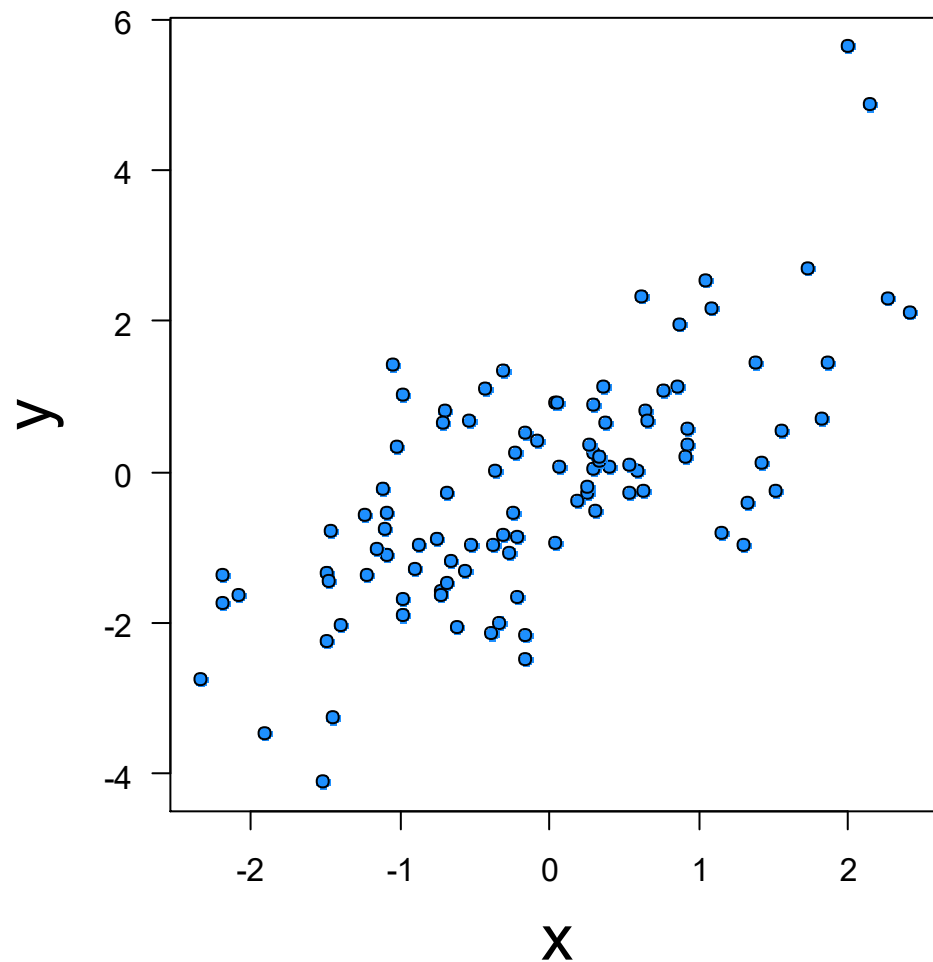
3. *Remember, these are based on probabilities and likelihoods. Sometimes, unlikely things happen (Type 1 error: false positives, Type 2 error: false negatives)*

3. Bayesian model selection

Let's start with a simple motivating example

(Script_7c_Bayesp_WAIC.R)

Data simulation

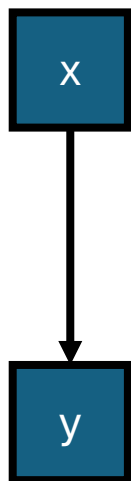


$$\mathbf{x} \sim \text{Normal}(0, 1)$$

$$\mathbf{y} \sim \text{Normal}(\mathbf{x}, 1)$$

$$n = 100$$

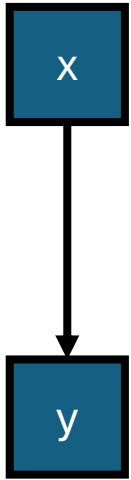
Run two models



$$\begin{aligned}y &\sim \text{Normal}(\beta x, \sigma^2) \\ \beta &\sim \text{Normal}(0, 1) \\ \sigma &\sim \text{Gamma}(1, 1)\end{aligned}$$

```
glm(y ~ 0 + x)
```

And calculate a bunch of stuff



glm

- The mean of β
- The p-value for β
- AICc

JAGS

- The median of β (q50)
- The $P(\beta > 0)$ (f-value)
- DIC (and pD)
- WAIC (and p_{WAIC})
- Bayesian p-value

I don't expect y'all to memorize these today!

There is explicit code to calculate these values, I do want to discuss it because it's useful

What is DIC?

What is DIC?

Deviance = -2 x log-likelihood

Effective number of parameters

$$\text{DIC} = \boxed{\bar{D}} + \boxed{pD}$$

$$\text{AICc} = \boxed{-2\log L} + \boxed{2k + \frac{2k^2 + 2k}{n - k - 1}}$$

Deviance = -2 x log-likelihood

Parameter penalty

They're very similar!

How do we get the log-likelihood?

$$\text{DIC} = \bar{D} + pD$$

$$pD = \left(\frac{\sigma_D^2}{2} \right)$$

We do this for every iteration

$$\log L = -\frac{n}{2} \times \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \times \sum (\mathbf{y} - \beta \mathbf{x})^2$$

$$D = -2 \times \log L$$

Well, JAGS (and Stan and Nimble) calculates it for us!

What is WAIC?

It's a lot like DIC and AICc with some improvements

We do this for every point at every iteration

$$\log L = -\frac{1}{2} \times \ln(2\pi) - \frac{1}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} \times (y - \beta x)^2$$

It's a lot like DIC and AICc with some improvements

- 1) We then take the mean of the likelihood (not log-likelihood) for each point across all iterations
- 2) We take the log of those means
- 3) And then we sum across points to get log pointwise predictive density (lppd)

It's a lot like DIC and AICc with some improvements

- 1) To get the effective number of parameters, we sum the variances of the log-likelihoods for each point

$$\text{WAIC} = -2 \times \text{lppd} + 2 \times p_{\text{WAIC}}$$

$$\text{DIC} = \bar{D} + pD$$

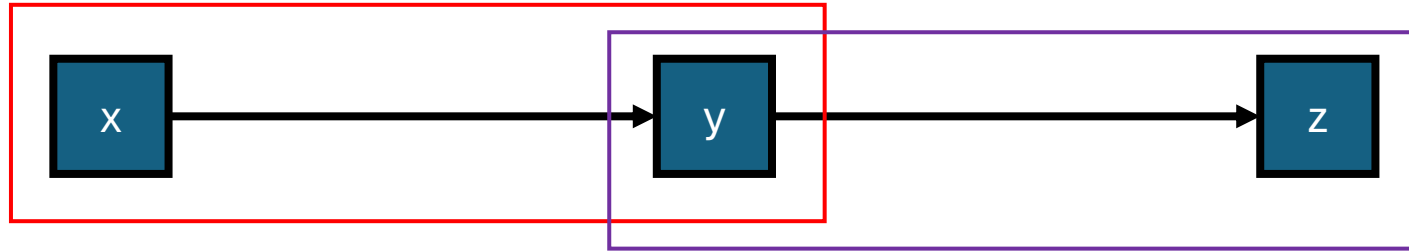
$$\text{AICc} = -2\log L + 2k + \frac{2k^2 + 2k}{n - k - 1}$$

DIC and WAICc (and BIC and others) are alternative metrics

- 1) For simple models, you won't find any differences
- 2) Different metrics can perform more effectively in certain circumstances
- 3) Conditional v. marginal approaches may vary when using models with latent variables

A couple of brief (and simple) segues

A really nice side bar about model selection in SEMs



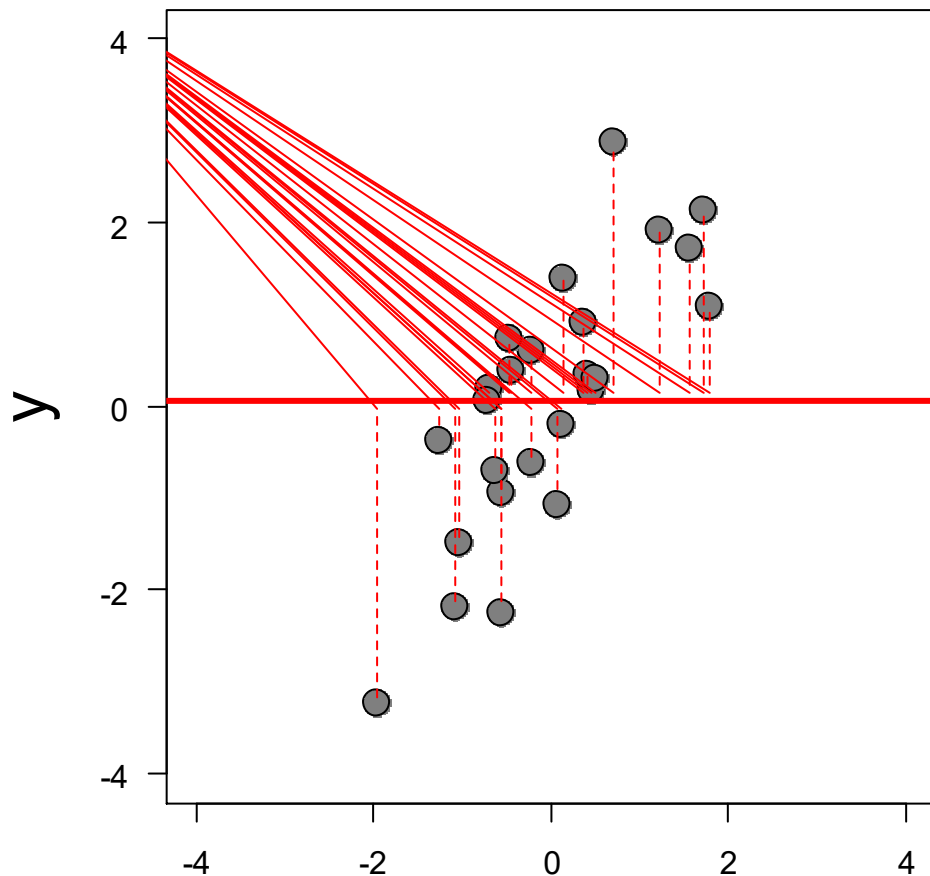
$$\text{WAIC for full model} = \sum \text{WAIC of sub-models}$$

What's up with 'effective parameters'?

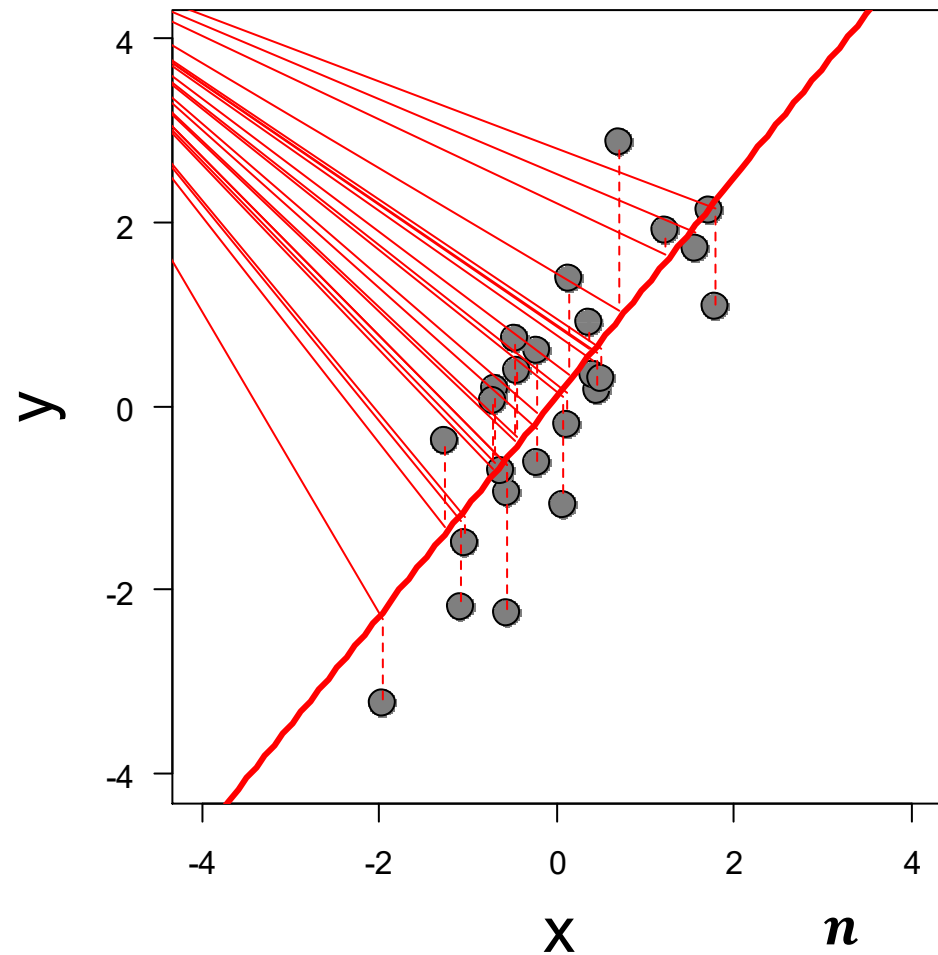
Imagine you use an incredibly strong prior that drives inference.
Should that count as a parameter?

4. What are Bayesian p-values?

Remember residual sums of squares?



$$50.156 = \sum_{i=1}^n (y_i - \eta_i)^2$$



$$19.448 = \sum_{i=1}^n (y_i - \eta_i)^2$$

What are Bayesian p-values?

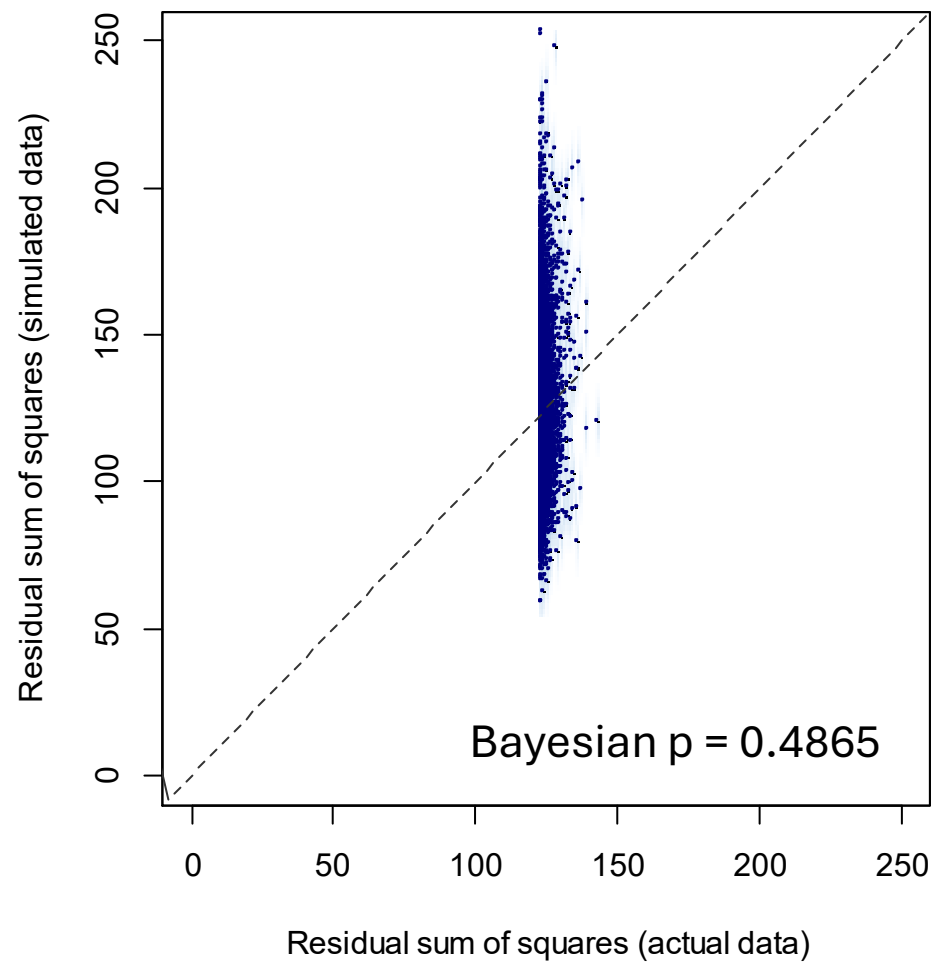
- We'll simulate new data (y') from our parameter estimates at every iteration

$$\mathbf{y} \sim \text{Normal}(\beta \mathbf{x}, \sigma^2)$$

$$\mathbf{y}' \sim \text{Normal}(\beta \mathbf{x}, \sigma^2)$$

- We'll calculate RSS for the new data (and the real data, note we often use other discrepancy statistics for other model types)
- We'll compare the RSS from simulated and real data. The p-value is the proportion of times that the RSS from the real data is greater than the RSS from the simulated data

What are Bayesian p-values?

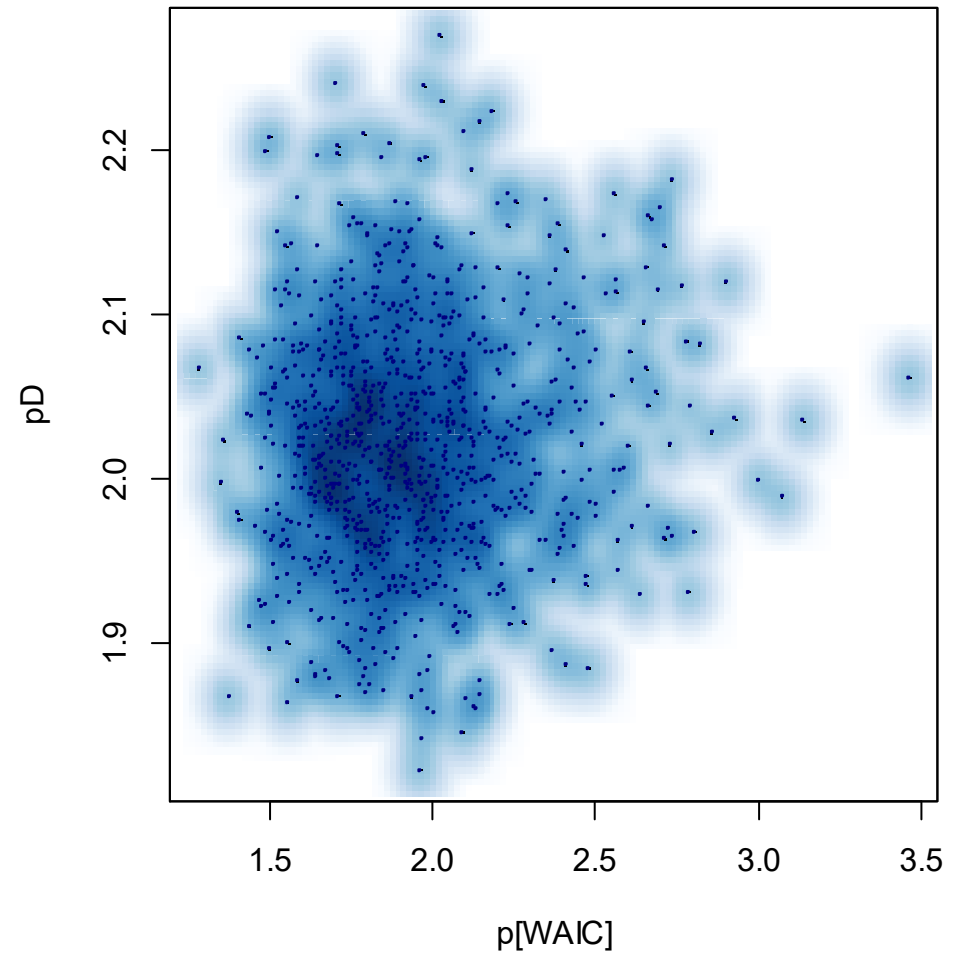
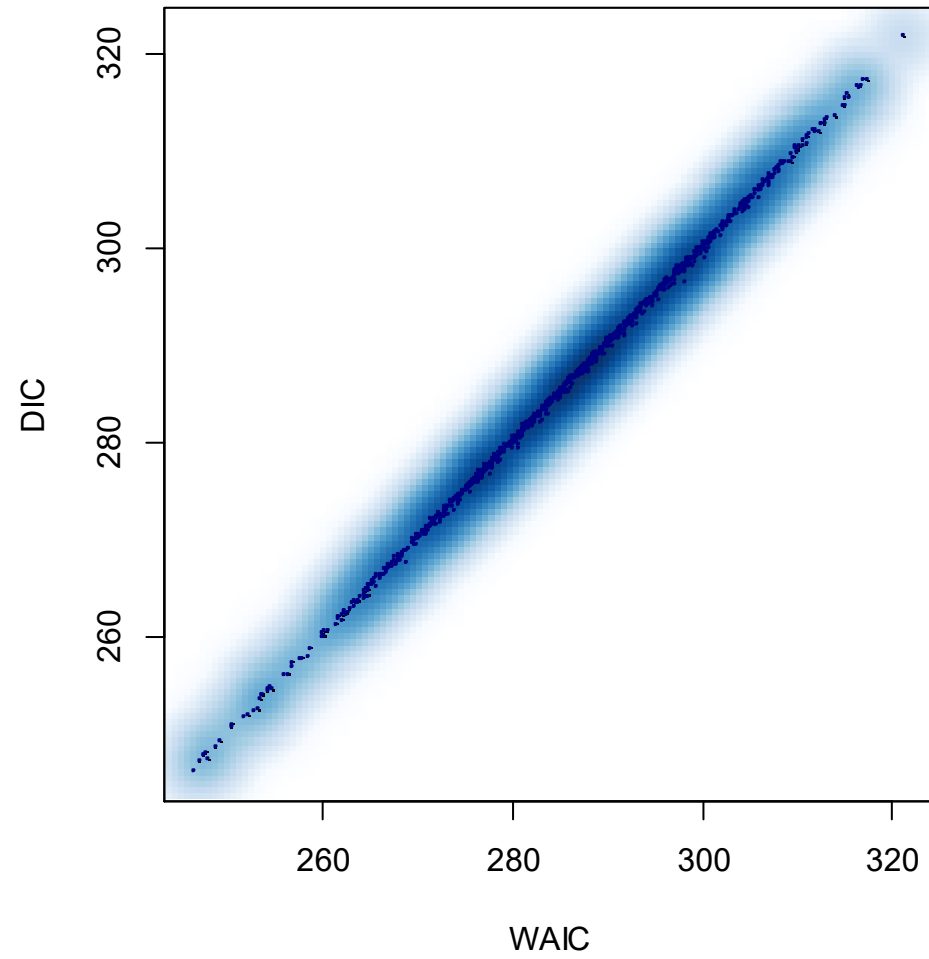


What are acceptable Bayesian p-values?

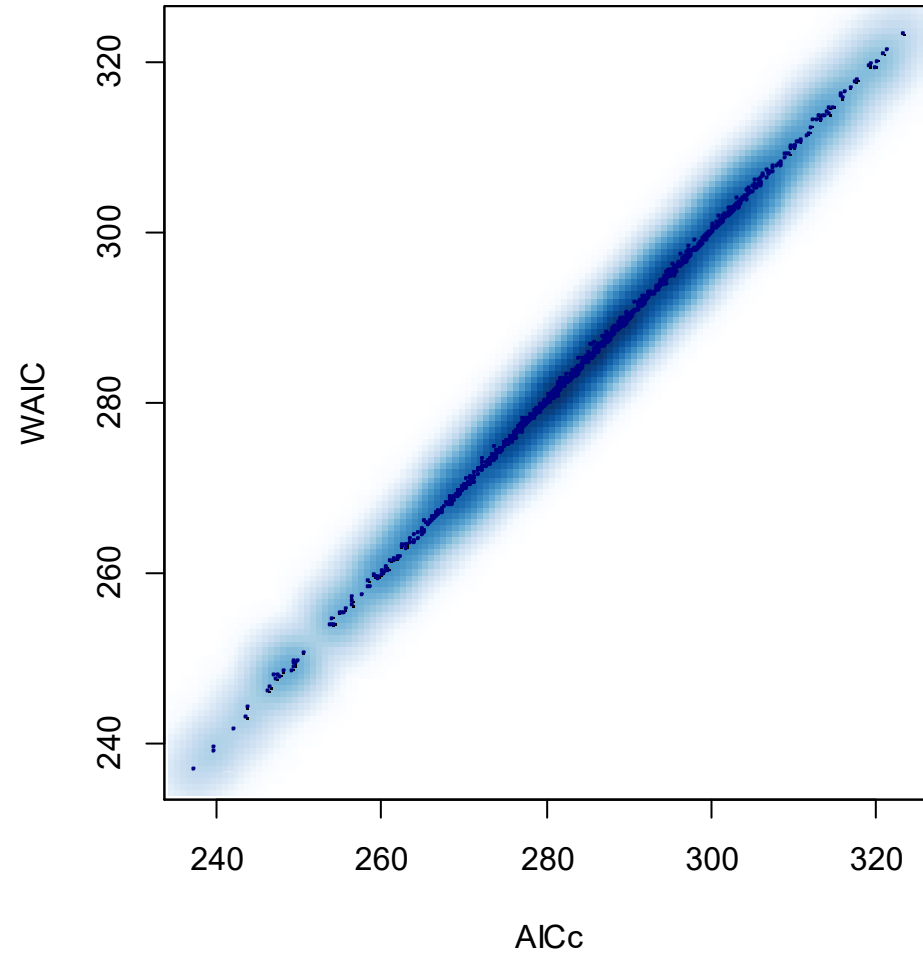
- A fitting model has a Bayesian p-value near 0.5, and values close to 0 or close to 1 suggest doubtful fit of the model (Kéry [2010] *Academic Press*)

Let's do all of that 1k times

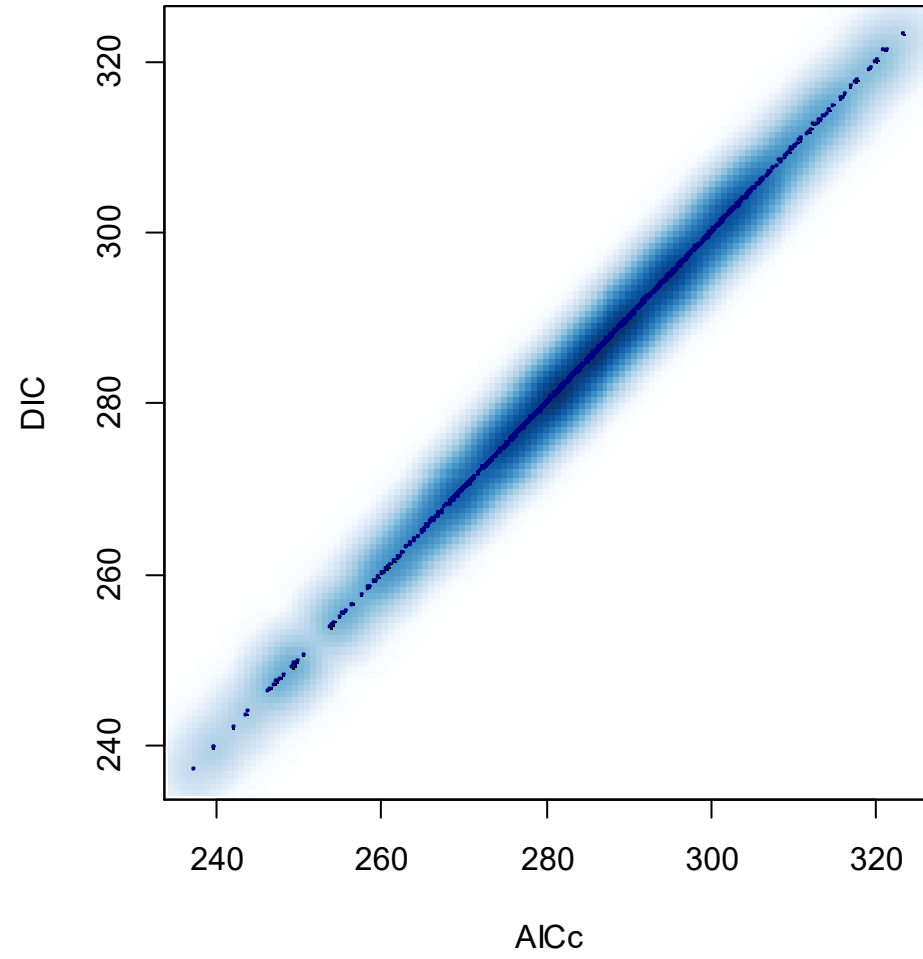
WAIC vs. DIC



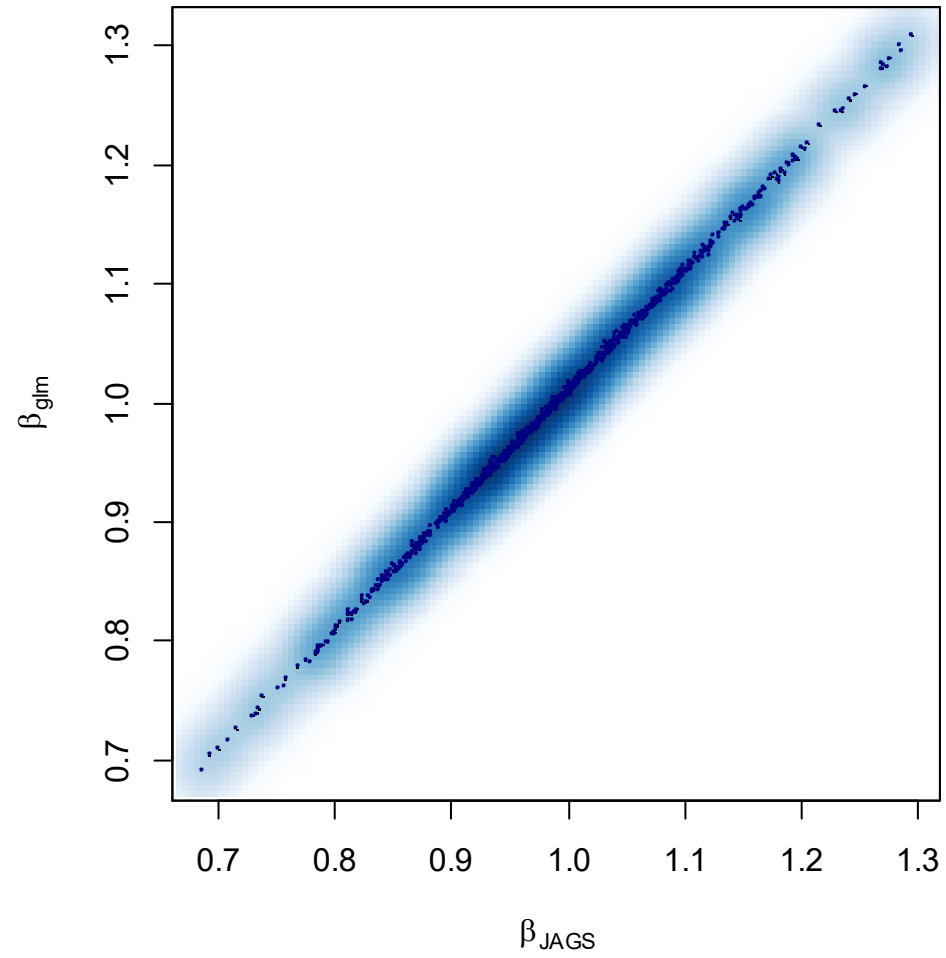
WAIC vs. AIC



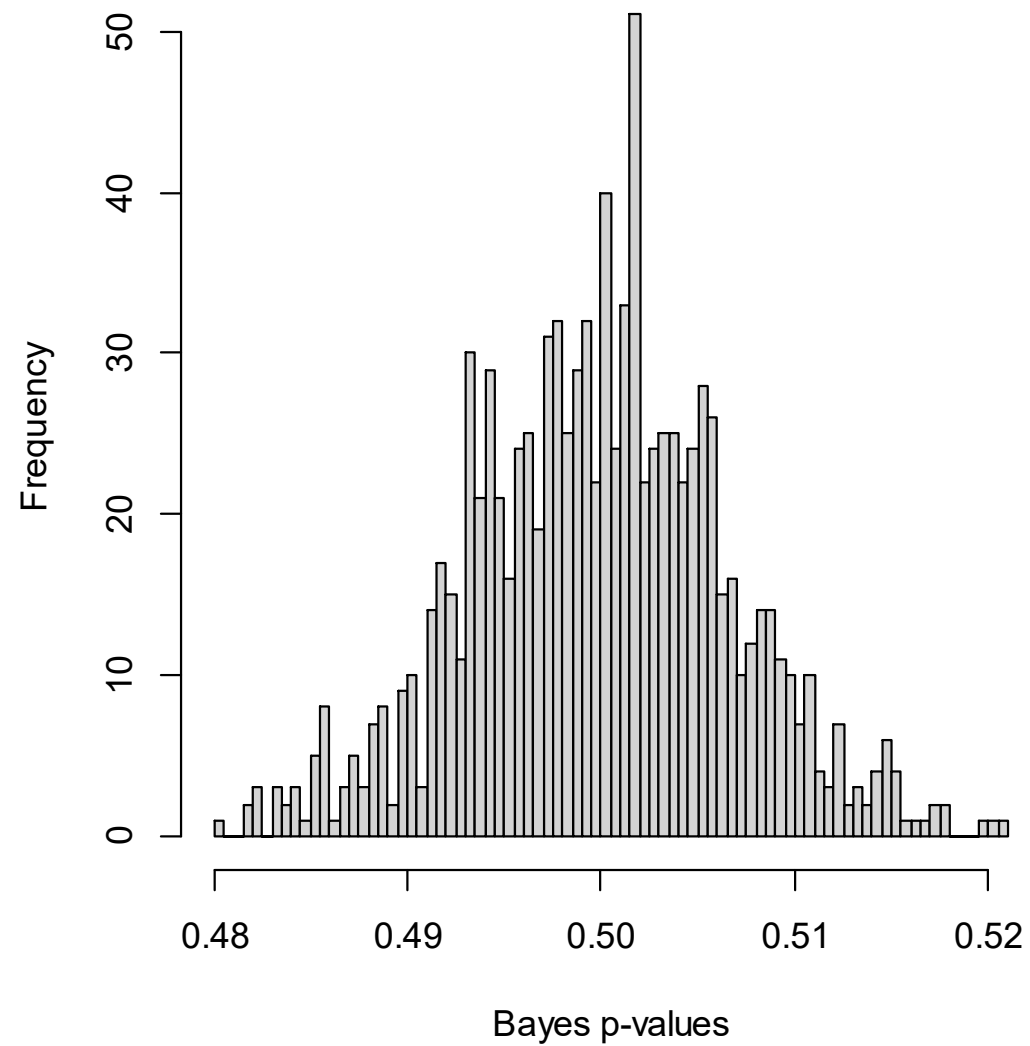
DIC vs. AIC



Parameter estimates



Bayesian p-values



For simple models, there will be no meaningful differences in model selection among DIC, WAIC, AICc, BIC, etc.

Ok, let's break some stuff again

Why not? It'll be fun

We're going to simulate clutch size as a function of body size

$$n = 200$$

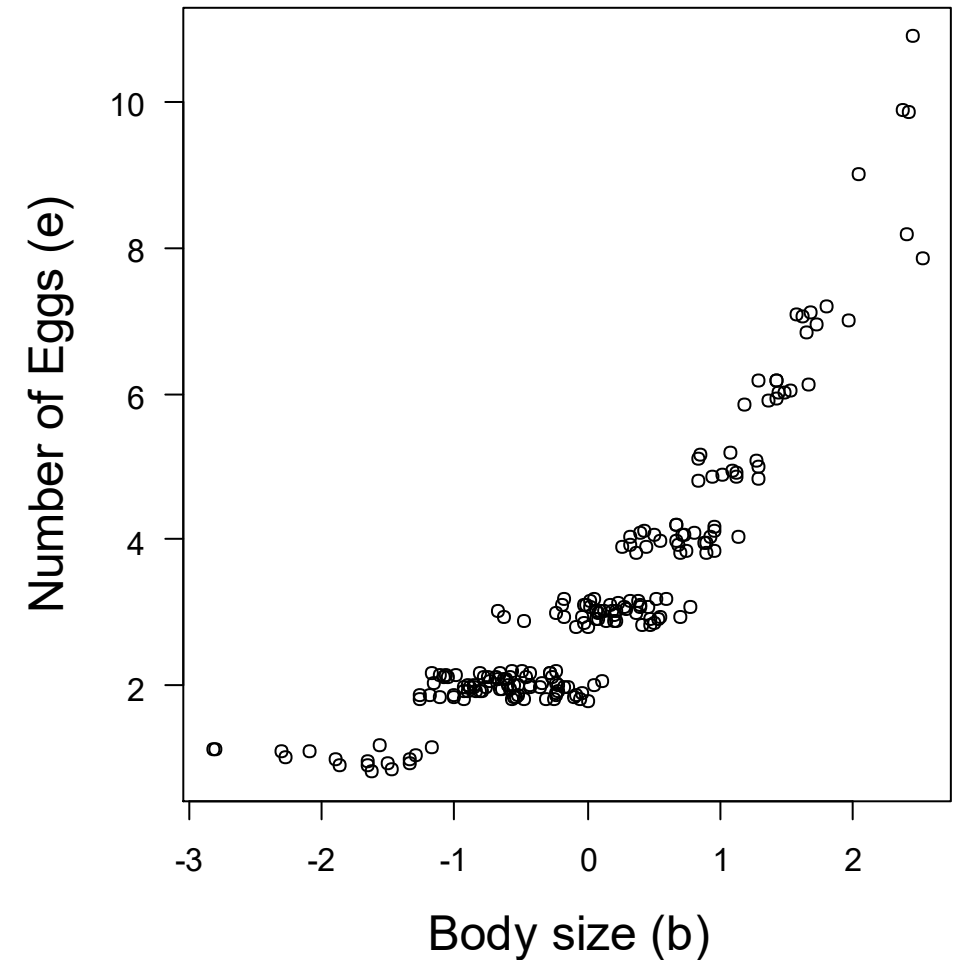
$$\mathbf{b} \sim \text{Normal}(0,1)$$

$$\psi = e^{\alpha + \beta \times \mathbf{b} + \text{noise}}$$

$$\mathbf{e} \sim \text{round}(\psi)$$

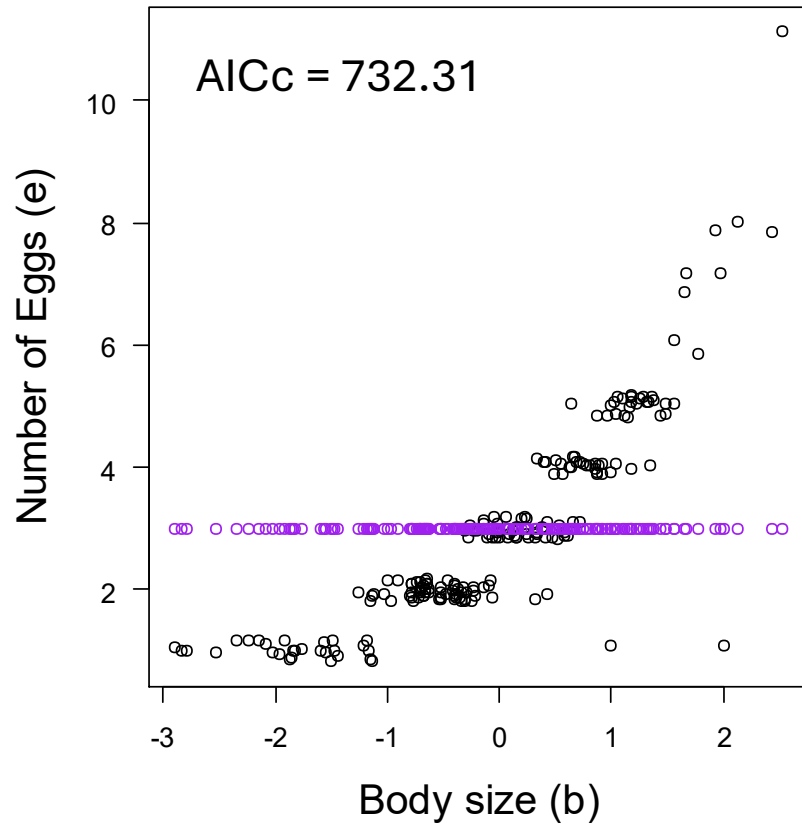
$$\alpha = 1$$

$$\beta = 0.5$$

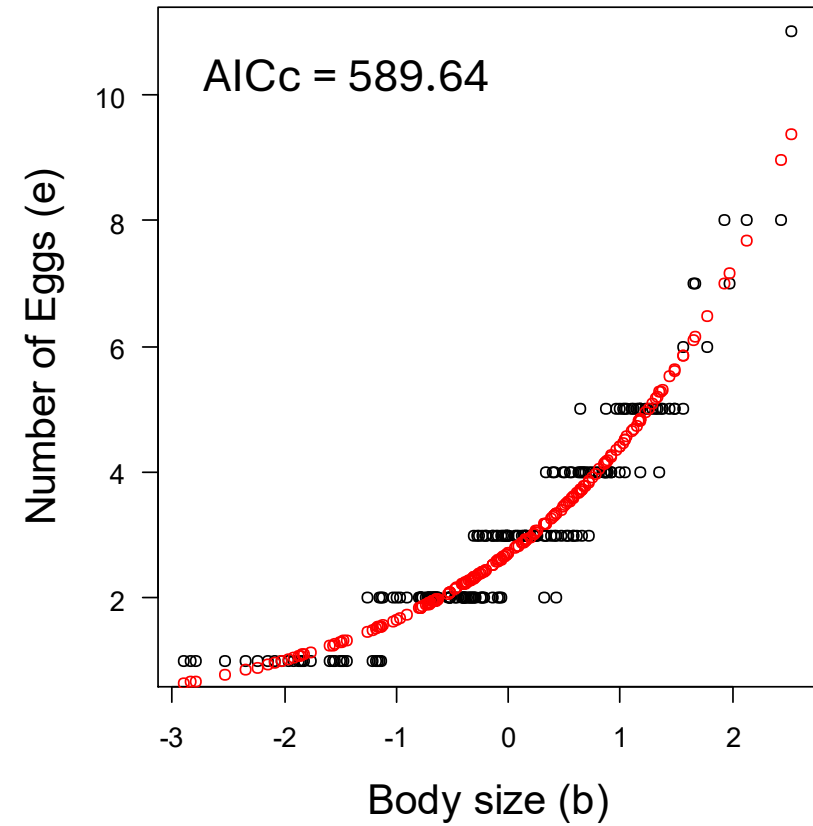


Residuals and AICc...

```
glm(y ~ 1, family = 'poisson')
```



```
glm(y ~ b, family = 'poisson')
```



Deviance (D) Information Criterion (DIC; run the same models in JAGS)

Model 0

$$\psi = e^{\alpha}$$

$$e \sim \text{Poisson}(\psi)$$

$$\alpha \sim \text{Normal}(1,1)$$

$$\text{DIC} = \bar{D} + 2pD$$

$$pD = \frac{\sigma_D^2}{2}$$

Model 1

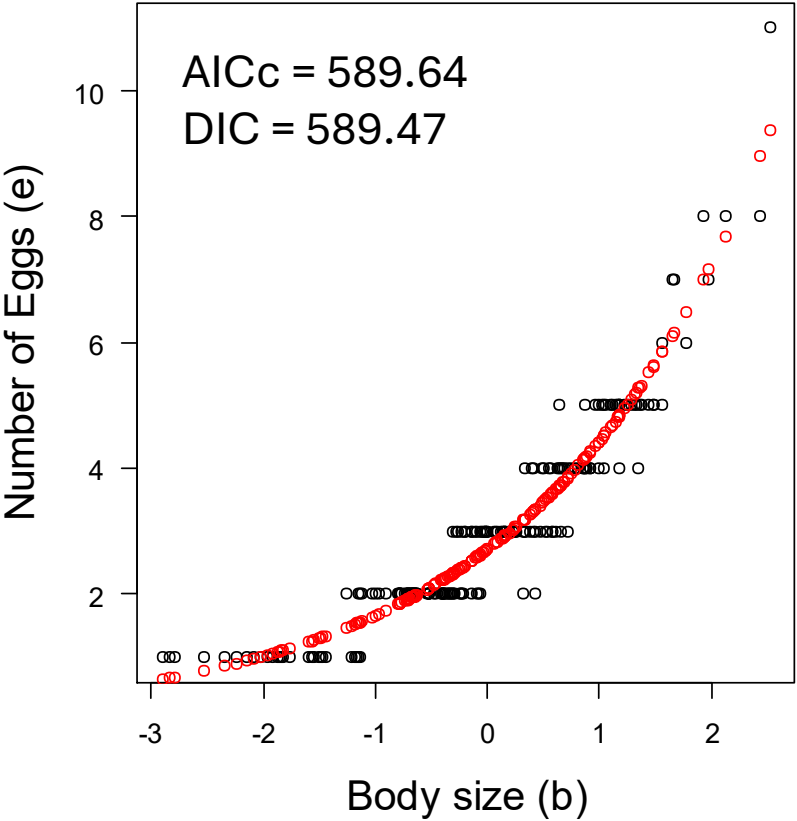
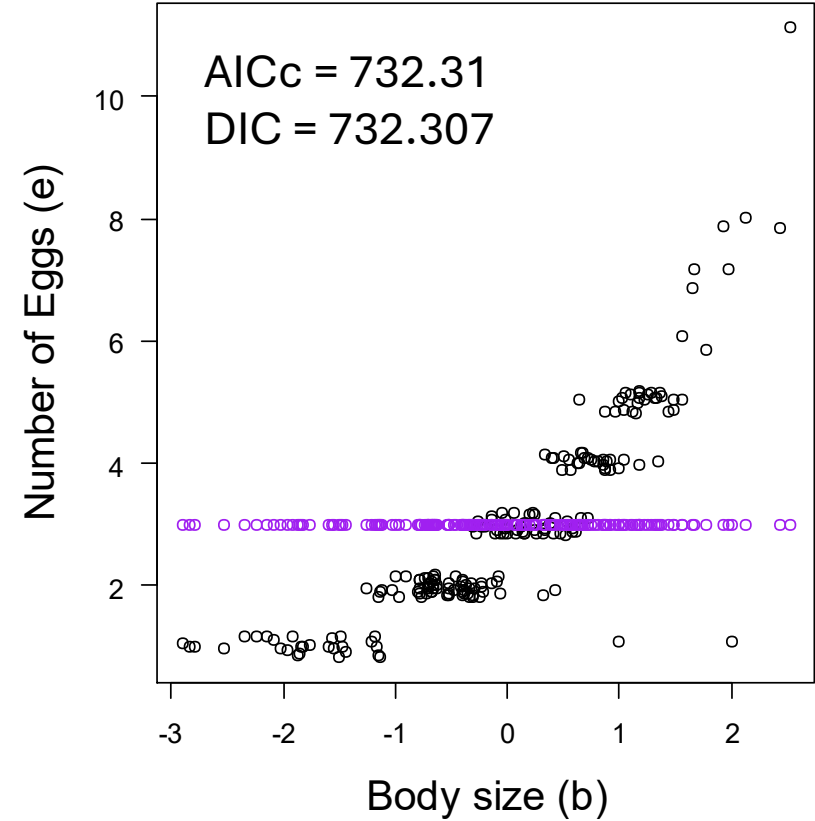
$$\psi = e^{\alpha + \beta \times b}$$

$$e \sim \text{Poisson}(\psi)$$

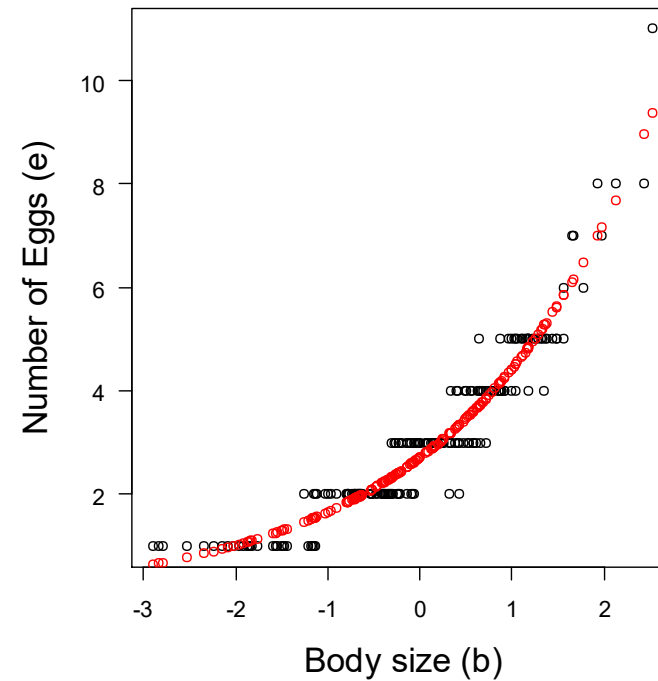
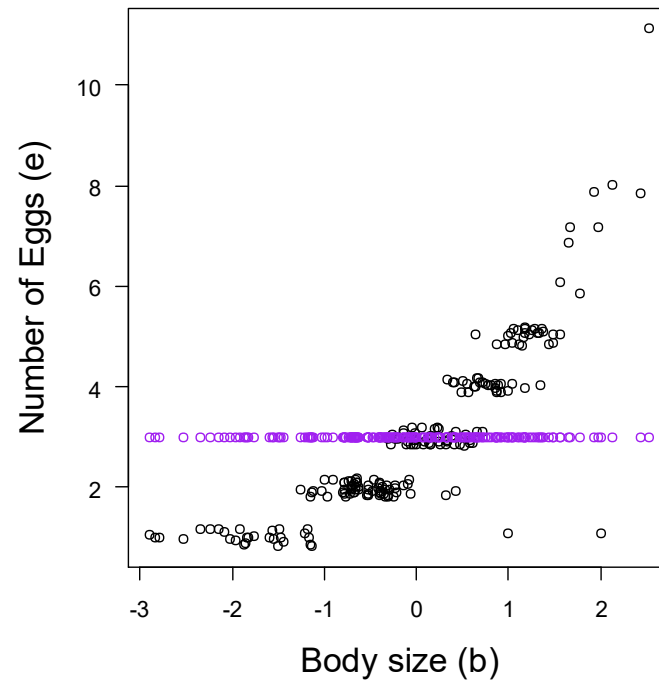
$$\alpha \sim \text{Normal}(1,1)$$

$$\beta \sim \text{Normal}(0,1)$$

DIC



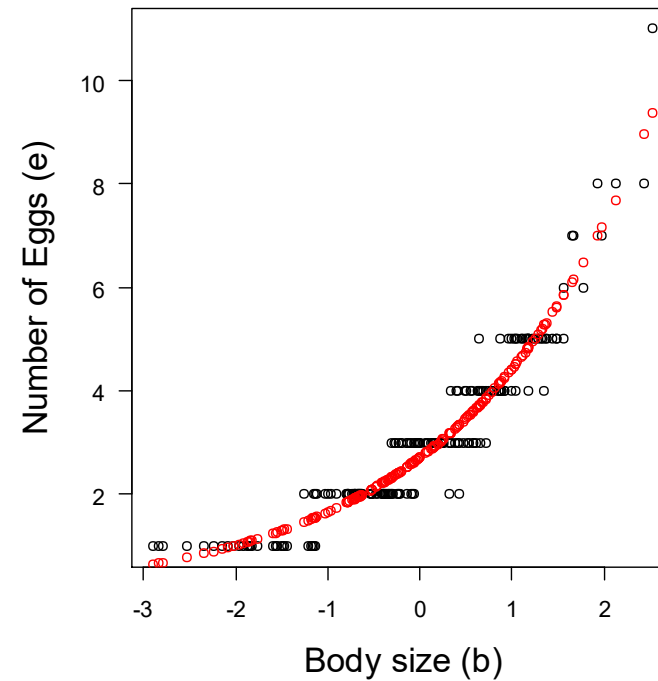
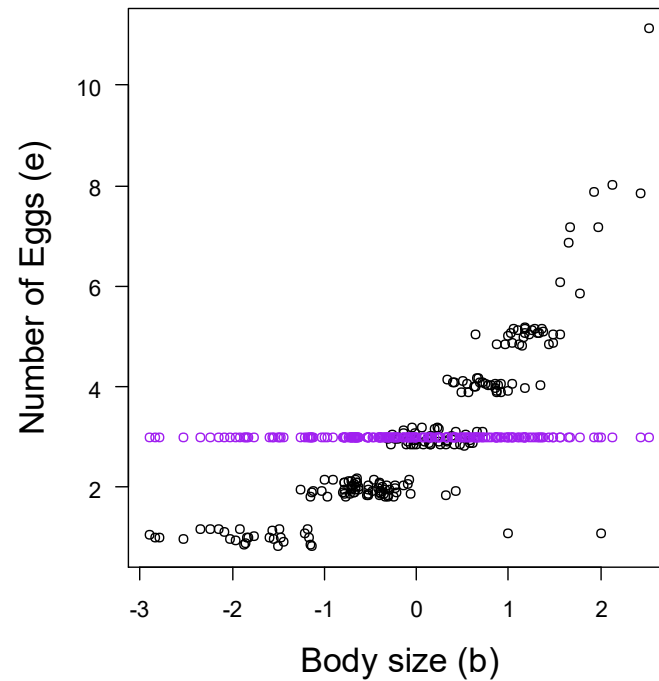
Bayesian p-values (RSS for this Poisson regression)



1. Calculate (Pearson's) residuals at each iteration

$$\varepsilon = \frac{e - e^{\beta_0 + \beta_1 \times b}}{\sigma_e}$$

Bayesian p-values (RSS for this Poisson regression)

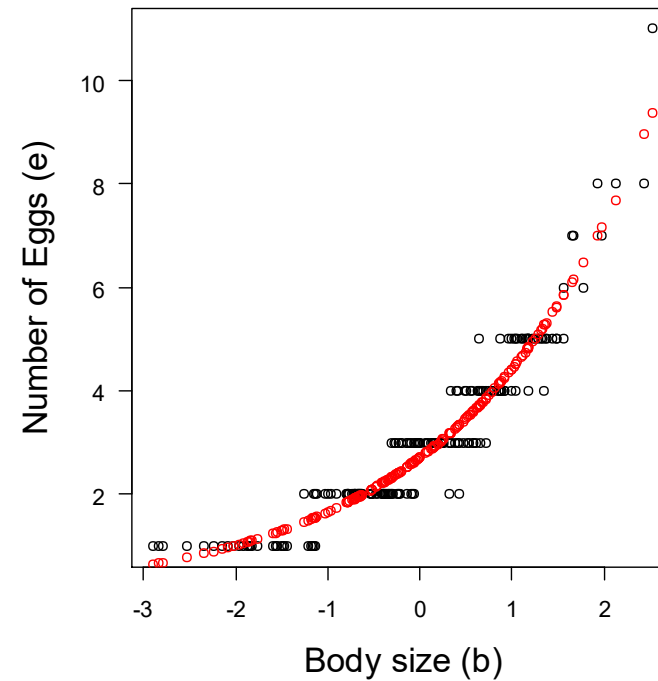
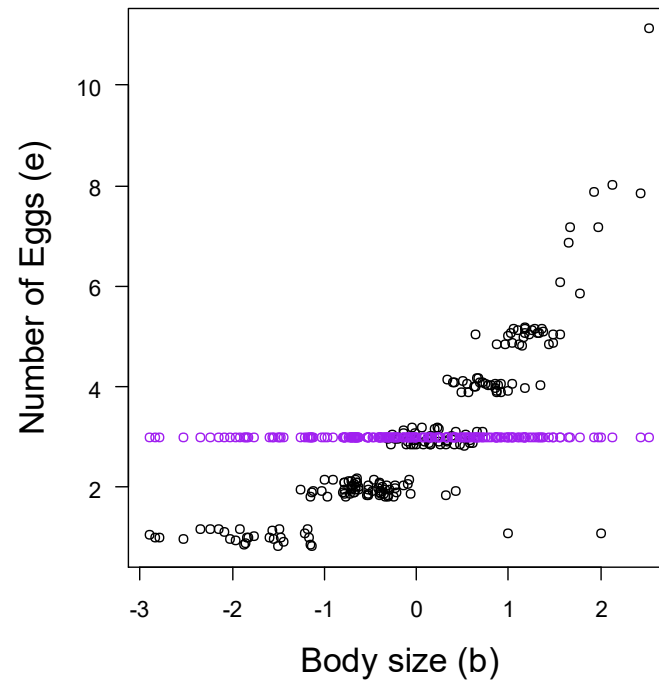


1. Calculate (Pearson's) residuals at each iteration
2. Generate 'new' data

$$e \sim \text{Poisson}(\psi)$$

$$e' \sim \text{Poisson}(\psi)$$

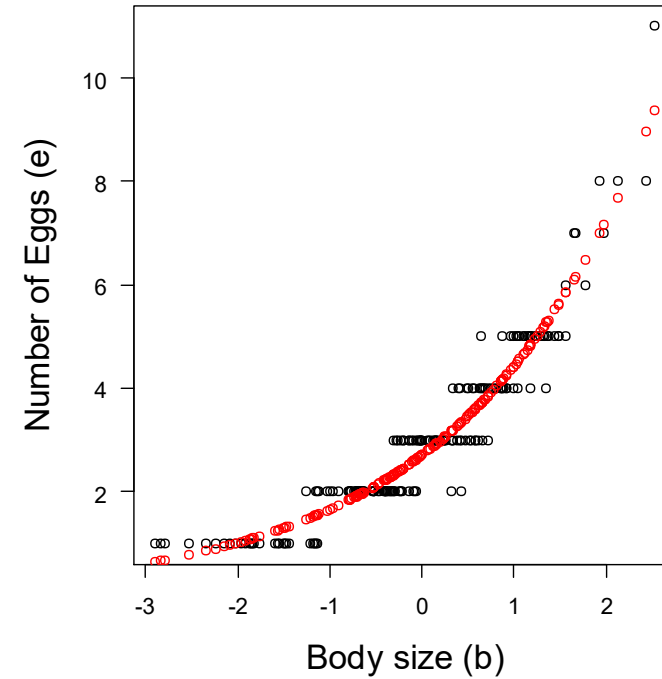
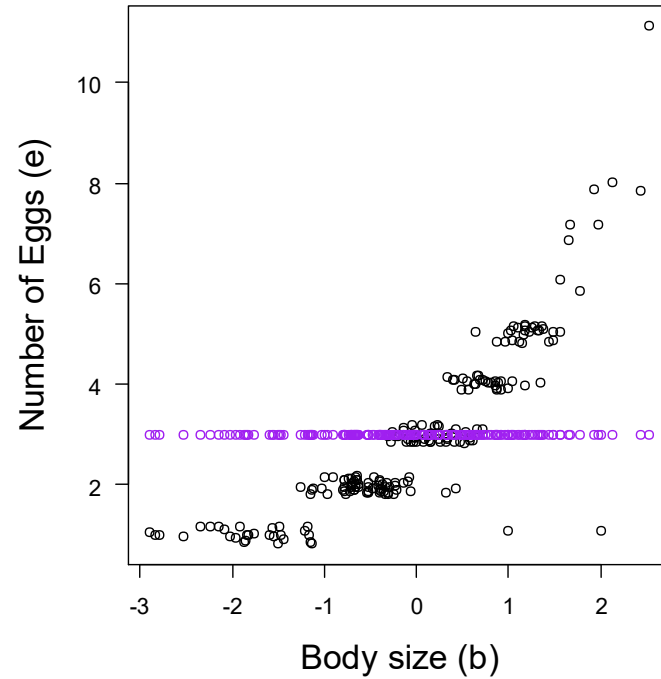
Bayesian p-values (RSS for this Poisson regression)



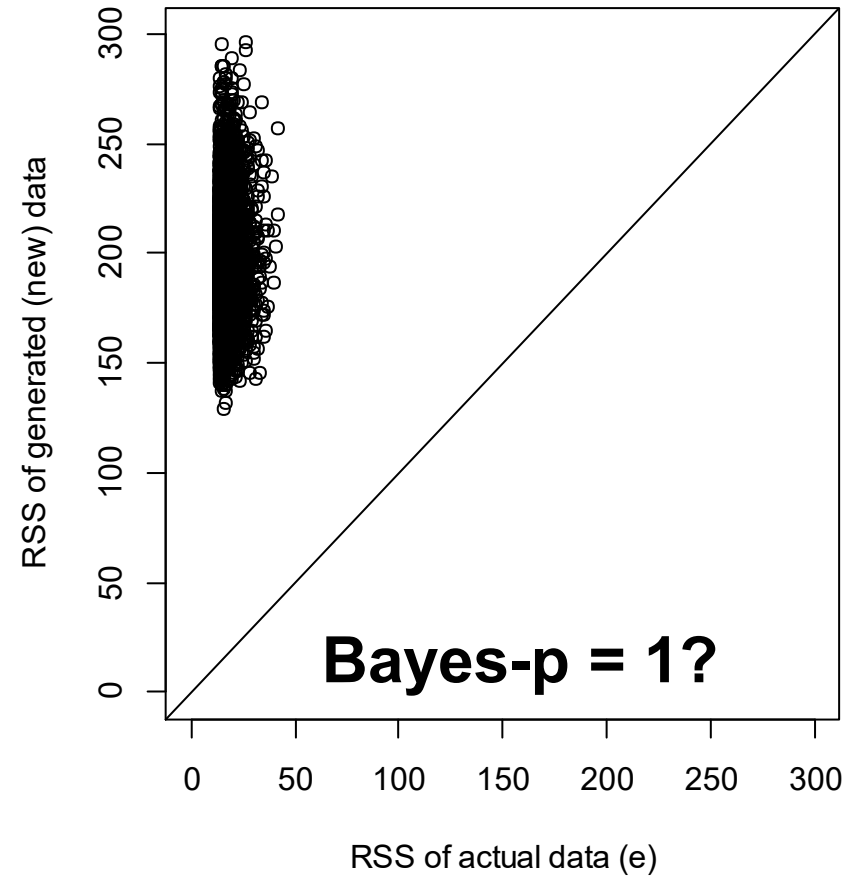
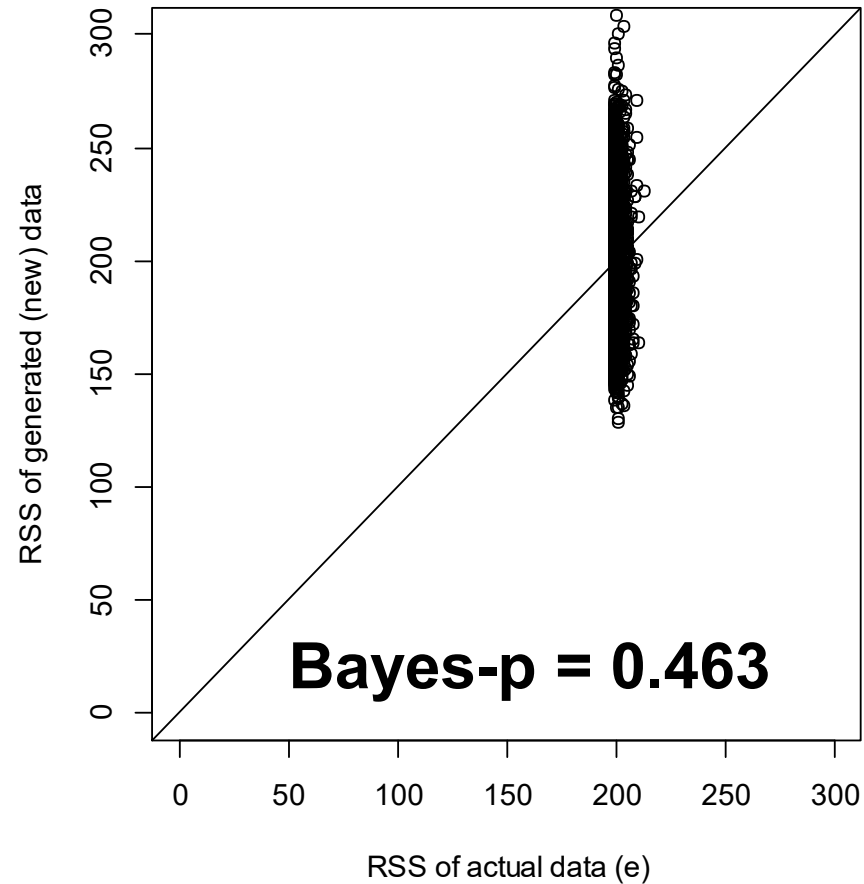
1. Calculate (Pearson's) residuals at each iteration
2. Generate 'new' data
3. Calculate (Pearson's) residuals at each iteration for new data

$$\varepsilon' = \frac{e' - e\beta_0 + \beta_1 \times b}{\sigma_{e'}}$$

Which model will fit the data better (have a Bayes p closer to 0.5)?

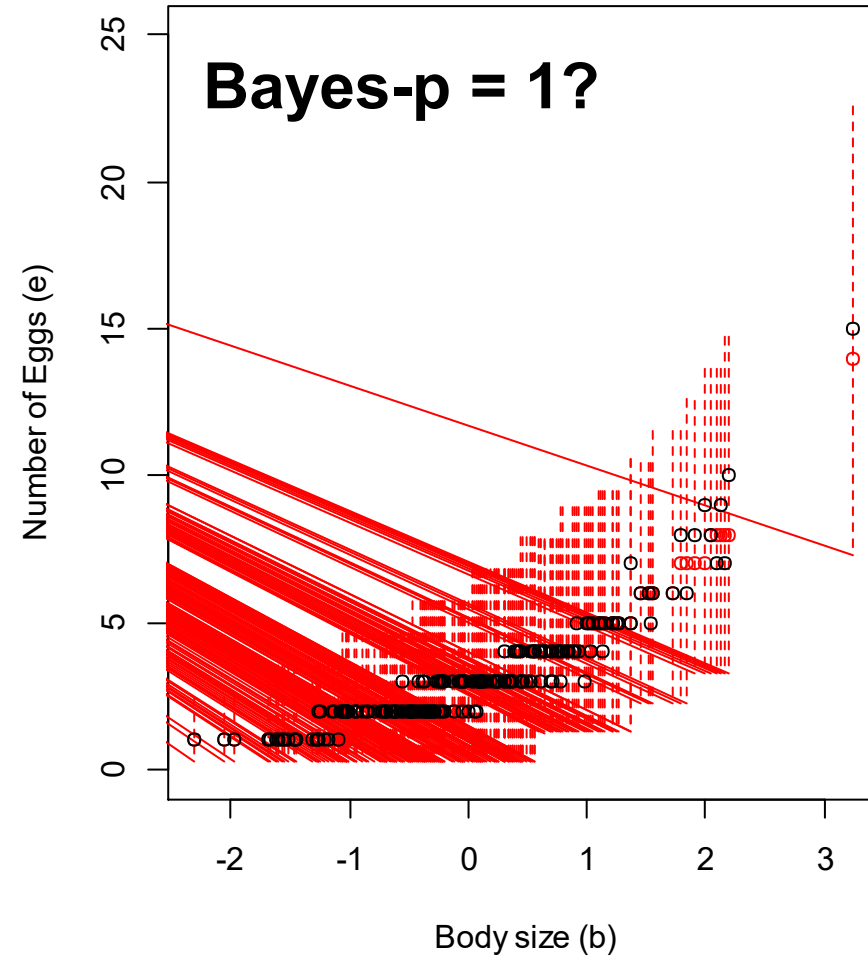
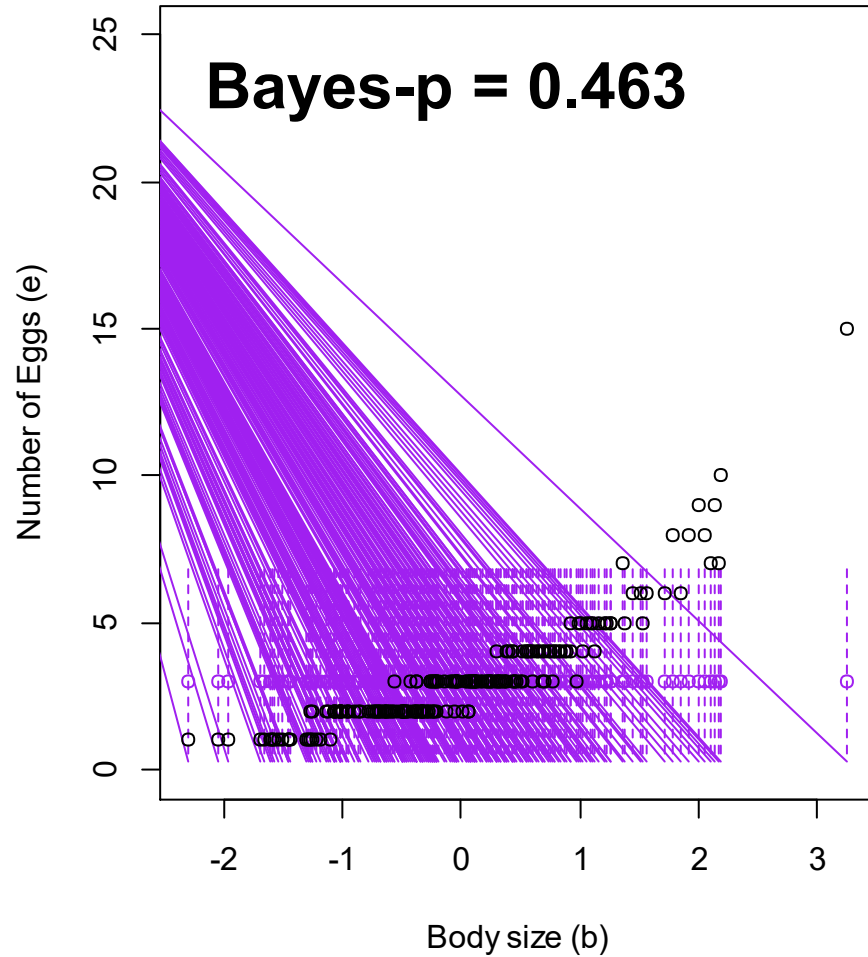


Oh for goodness sake!!



Bayes-p > 0.5 means real data fit better than simulated, and vice versa

Oh for goodness sake!!



Bayes-p > 0.5 means real data fit better than simulated, and vice versa

Let's dig into the code and find out why!?

It will be fun.

A take-home message here is that poor model fit doesn't necessarily mean a 'bad' model. Rather, it may just mean a bit more thought is necessary for addressing error?

